NSSECU3 Digital Forensics AY2526T2

Laboratory Activity Report (LAR) / Digital Forensics Report (DFR) / Mini Project (MP) / Write-Up (WUP) / Windows Forensics Project (WFP)

MAGNET VIRTUAL SUMMIT 2026 CTF WRITE-UP

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Approved By:	Jerome E. Gutierrez NSSECU3 Course Instructor
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Submitted By:	Juan Pedro B. Cruz
Group Members:	Wendel Walter A. Leander, Kyle Gabriel A. Maristela, Owen Phillip C. San Luis, John Christian N. Sayat

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Task Distribution Summary

Description	Summarizes all the checklist activities given by the instructor as well as the specific tasks each student was assigned and checked by their groupmates approval.
Start of Activity	27 Feb 2026
End of Activity	09 Mar 2026
Main Contributor	All

Change Control

Version	Changes	Date	Changed by
1.0	Fixed template and drafted sections	5 March 26	Wendel Walter A. Leander
1.1	Drafted purpose, scope, definitions, and Android Analysis	6 March 26	Wendel Walter A. Leander
1.2	Added IOS Analysis	6 March 26	John Christian N. Sayat
1.3	Added Cipher Analysis	6 March 26	Owen Phillip C. San Luis
1.4	Updated Task Lists	7 March 26	Wendel Walter A. Leander

Task List

#	Item	Purpose	Assignee	Status
1	Evidence Collection and Preparation	To download the provided CTF images for extraction and analysis.	All	Done
2	Tool Setup	To prepare the tools for artifact analysis and flag identification.	All	Done
3	Initial Data Parsing	To convert raw, unstructured, or proprietary data extracted from mobile images into human-readable information.	All	Done
4	Artifact Analysis	To reconstruct user activity, establish timelines, and uncover hidden evidence from mobile device dumps	All	Done
5	Flag Identification	To provide concrete and verified proof that a flag has been successfully identified.	All	Done

Member's Task Lists

Detailed Task List by Wendel Leander

#	Item	Purpose	Appendix	Status
1	Collection of CTF Images and Files	To download the provided CTF images for extraction and analysis.	Appendix A	Done
2	Flag Identification and Image Analysis	To identify the flags required by each challenge on the Android and IOS images as well as the Ciphers.	Appendix A & B	Done
3	Tools Used	To list the tools used in the identification of flags, preprocessing of images, and parsing of files and data.	Appendix B	Done
4	MVS 2026 CTF Group Rankings	To show the final rankings of the group in the Magnet Virtual Summit 2026 CTF.	Appendix C	Done

Detailed Task List by Kyle Maristela

#	Item	Purpose	Appendix	Status
1	Collection of CTF Images and Files	To download the provided CTF images for extraction and analysis.	Appendix A	Done
2	Flag Identification and Image Analysis	To identify the flags required by each challenge on the Android and IOS images as well as the Ciphers.	Appendix A & B	Done
3	Tools Used	To list the tools used in the identification of flags, preprocessing of images, and parsing of files and data.	Appendix B	Done
4	MVS 2026 CTF Group Rankings	To show the final rankings of the group in the Magnet Virtual Summit 2026 CTF.	Appendix C	Done

Detailed Task List by Owen San Luis

#	Item	Purpose	Appendix	Status
1	Collection of CTF Images and Files	To download the provided CTF images for extraction and analysis.	Appendix A	Done
2	Flag Identification and Image Analysis	To identify the flags required by each challenge on the Android and IOS images as well as the Ciphers.	Appendix A & B	Done
3	Tools Used	To list the tools used in the identification of flags, preprocessing of images, and parsing of files and data.	Appendix B	Done

4	MVS 2026 CTF Group Rankings	To show the final rankings of the group in the Magnet Virtual Summit 2026 CTF.	Appendix C	Done

Detailed Task List by John Sayat

#	Item	Purpose	Appendix	Status
1	Collection of CTF Images and Files	To download the provided CTF images for extraction and analysis.	Appendix A	Done
2	Flag Identification and Image Analysis	To identify the flags required by each challenge on the Android and IOS images as well as the Ciphers.	Appendix A & B	Done
3	Tools Used	To list the tools used in the identification of flags, preprocessing of images, and parsing of files and data.	Appendix B	Done
4	MVS 2026 CTF Group Rankings	To show the final rankings of the group in the Magnet Virtual Summit 2026 CTF.	Appendix C	Done

Purpose

The primary objective of this activity is to conduct forensic analysis on mobile device artifacts to identify hidden flags presented in the Magnet 2026 Digital Forensics CTF challenges. This activity allows participants to analyze digital artifacts, solve challenging ciphers, and identify hidden flags across multiple forensic scenarios involving mobile devices, application data, and encoded information.

On a practical level, the activity is intended to reinforce investigative thinking and the use of forensic tools when examining real-world style artifacts. This activity aims to gain hands-on experience navigating Android and iOS application data, analyzing SQLite databases, interpreting system configuration files, figuring out cipher challenges, and performing data decoding techniques to locate challenge flags. Although participation in the tournament was optional, it served as a supplementary exercise designed to familiarize students with Capture-the-Flag (CTF) style forensic challenges commonly encountered in professional digital forensics competitions and training environments.

Scope

This document focuses on the forensic analysis performed during the Magnet Digital Forensics Tournament 2026 challenges completed as part of the optional training activity. The scope includes the investigation of Android application artifacts, SQLite databases, system configuration files, and encoded challenge data in order to identify challenge flags. Challenges included Android, iOS, and cryptography-based tasks that required artifact examination and logical reconstruction of user activity.

The forensic analysis primarily involved examining mobile application data such as configuration files, databases, and logs. Tools used during the activity included forensic utilities such as ALEAPP for Android artifact parsing and DB Browser for SQLite for manual database inspection. Additional command-line tools and scripting methods were used to convert timestamps, search through extracted files, and analyze encoded values when necessary.

This document does not include reverse engineering of mobile applications or modification of application binaries. The scope is limited to artifact analysis, interpretation of recovered data, and the documentation of investigative steps used to identify challenge flags within the provided datasets.

Definitions and Acronyms

The technical terms, abbreviations, and jargon used in this document are listed below in ascending alphabetical order.

- Artifact - Digital evidence extracted from devices, such as application databases, logs, configuration files, and system settings that reflect user activity.
- Capture The Flag (CTF) - A cybersecurity challenge format where participants analyze provided data or systems to discover hidden values known as flags.
- CTF Flag - A specific piece of information or value discovered during a challenge that confirms successful completion of a forensic task.
- Cipher - A cryptographic algorithm used to transform readable data (plaintext) into an encoded format (ciphertext) to protect information from unauthorized access.
- Digital Artifact - Data generated by user actions or system processes that can provide evidence of activity during forensic analysis.
- Forensic Tool - Software used to examine and interpret digital evidence while maintaining the integrity of the data.
- SQLite Database - A lightweight relational database format commonly used by mobile applications to store structured data such as messages, logs, and user settings.
- Timestamp - A numeric or formatted value representing the date and time an event occurred within a digital system.
- UTC (Coordinated Universal Time) - The global time standard used in digital systems for recording and comparing timestamps across different time zones.
- Mobile Forensics - The process of extracting and analyzing data from mobile devices to reconstruct user activity and identify relevant evidence.

Roles and Responsibilities

The students involved in the process of the second laboratory activity are as listed below:

1. Leander, Wendel Walter A.
    a. Activity Report
    b. CTF Participation
2. Maristela, Kyle Gabriel A.
    a. Activity Report

3. San Luis, Owen Philip C.
    a. Activity Report
    b. CTF Participation
4. Sayat, John Christian N.
    a. Activity Report
    b. CTF Participation

Materials and Equipment

The software and tools required to perform the procedure are as listed below:

1. Software for Android Analysis
    a. DB Browser (SQLite)
    b. ALEAPP
2. Software for IOS Analysis
    a. DB Browser (SQLite)
    b. WinRar
    c. Google Maps
3. Software for Cipher Analysis
    a. Pixlr
    b. FTK Imager
    c. DeepSound
    d. Magnet Axiom

Procedure Steps

This document focuses on the forensic analysis performed during the Magnet Digital Forensics Tournament 2026 challenges completed as part of the optional training activity. The scope includes the investigation of Android application artifacts, SQLite databases, system configuration files, and encoded challenge data in order to identify challenge flags. Challenges included Android, iOS, and cryptography-based tasks that required artifact examination and logical reconstruction of user activity.

Android Challenges

Anything open this time of night? (5 pts)

Question: What is the name of the establishment the user took a screenshot of?

Flag was actually accidentally discovered when exploring the files, an image in the screenshot folder shows a website with a restaurant name and a time of 3:38 which one would assume is in early morning, meeting the requirements of the question.

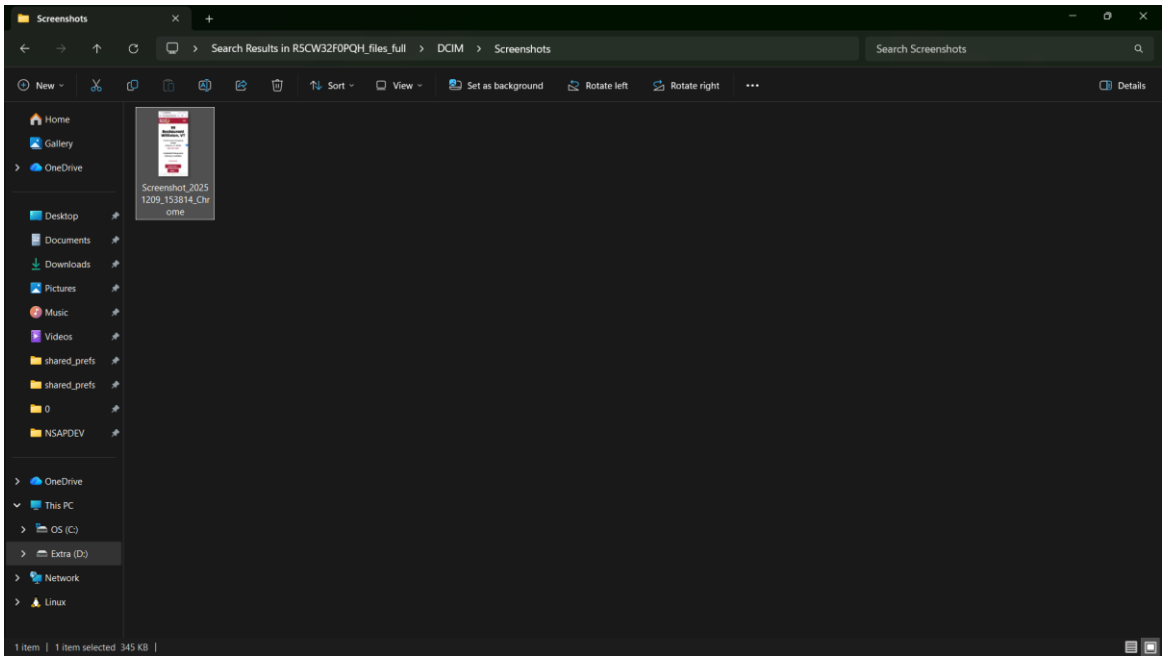


Figure x: Location of Screenshot in the Files

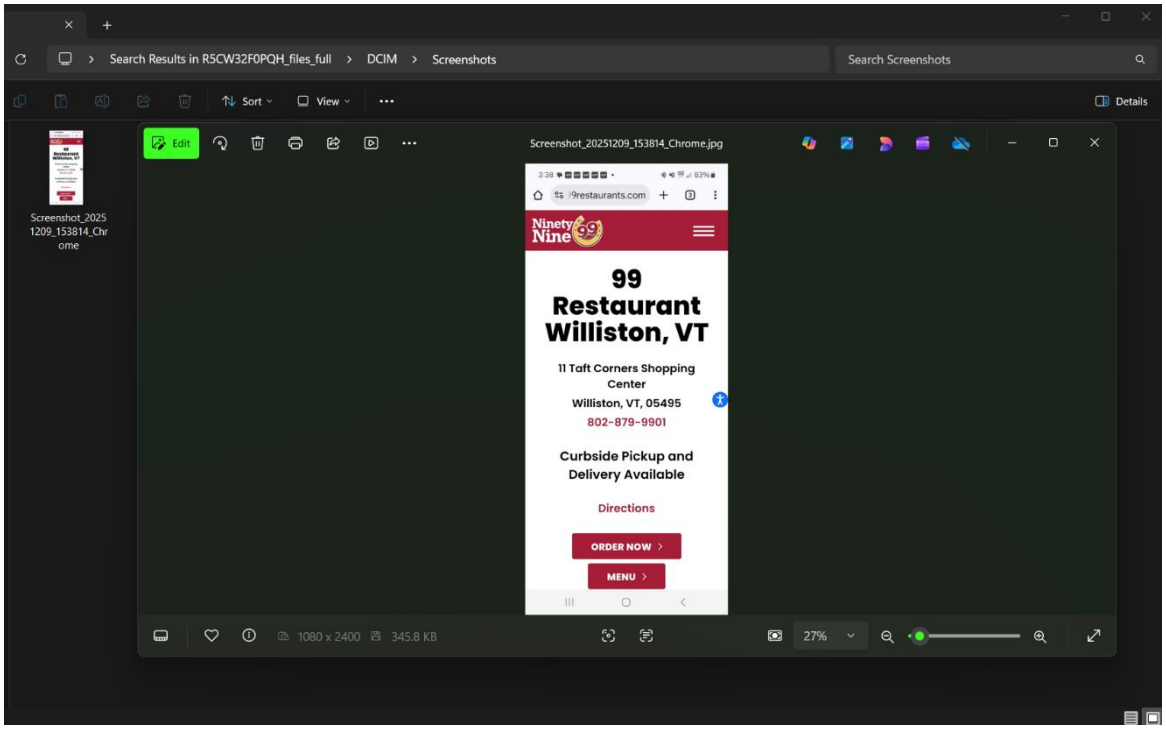


Figure x: Screenshot Image showing the Flag

Flag: 99 Restaurant

Three Star Exit (5 pts)

Question: What was the last time in UTC Clash Of Clans was closed? (Format: YYYY-MM-DD HH:MM:SS)

The Samsung A53 image was analyzed using ALEAPP v3.4.0 to extract Android forensic artifacts and generate a report. In the ALEAPP report, the Samsung Digital Wellbeing - Events section was examined and filtered for the package com.supercell.clashofclans.

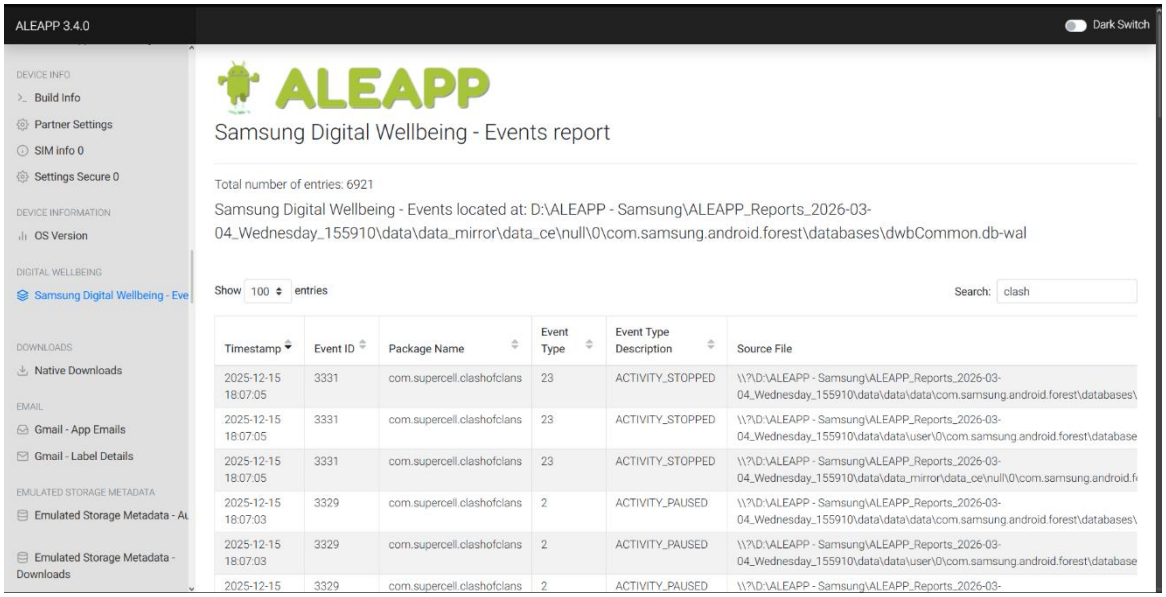


Figure x: Samsung Digital Wellbeing – Events report via ALEAPP

The event ACTIVITY_STOPPED indicates when the Clash of Clans application was closed. The latest event showed the timestamp 2025-12-15 18:07:05, which was submitted as the flag.

Flag: 2025-12-15 18:07:05

Long Distance Call (5 pts)

Question: What country was the other party from for a rejected/declined call?

The Samsung A53 image was analyzed using ALEAPP v3.4.0 to extract Android forensic artifacts and generate a report. In the ALEAPP report, the Call Logs section was examined to review all recorded phone calls.

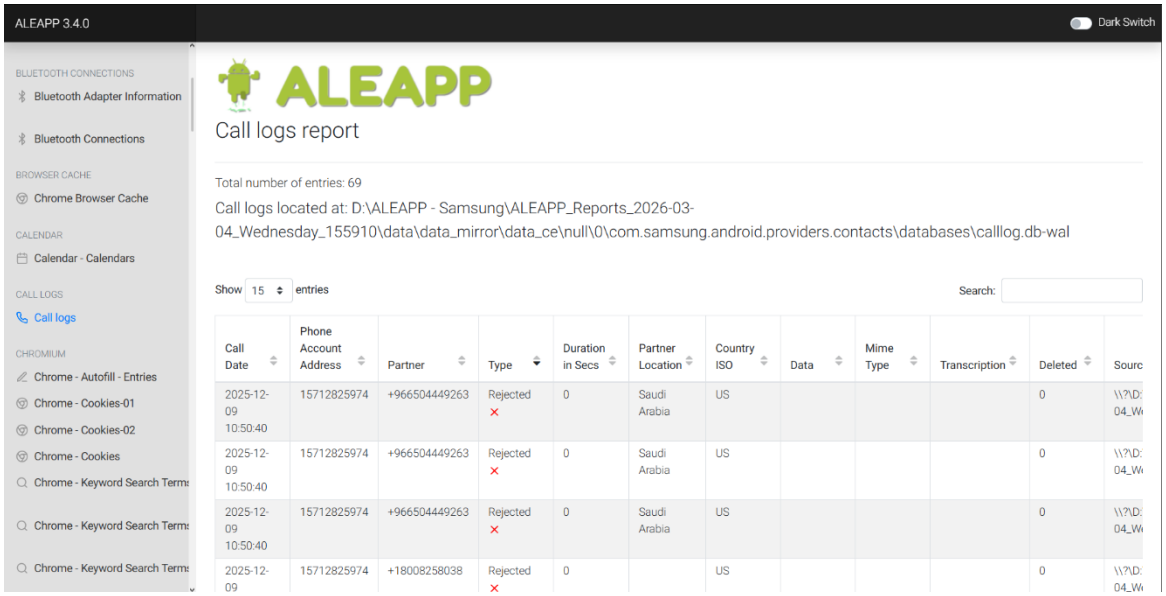


Figure x: Call logs report via ALEAPP

Clean Lens (10 pts)

Question: What is the Ingress rating of the security camera that supports night vision?

The Samsung A53 image was analyzed using ALEAPP v3.4.0 to extract Android forensic artifacts and generate a report. In the ALEAPP report, the Chrome Web History section was reviewed to identify websites visited by the user.

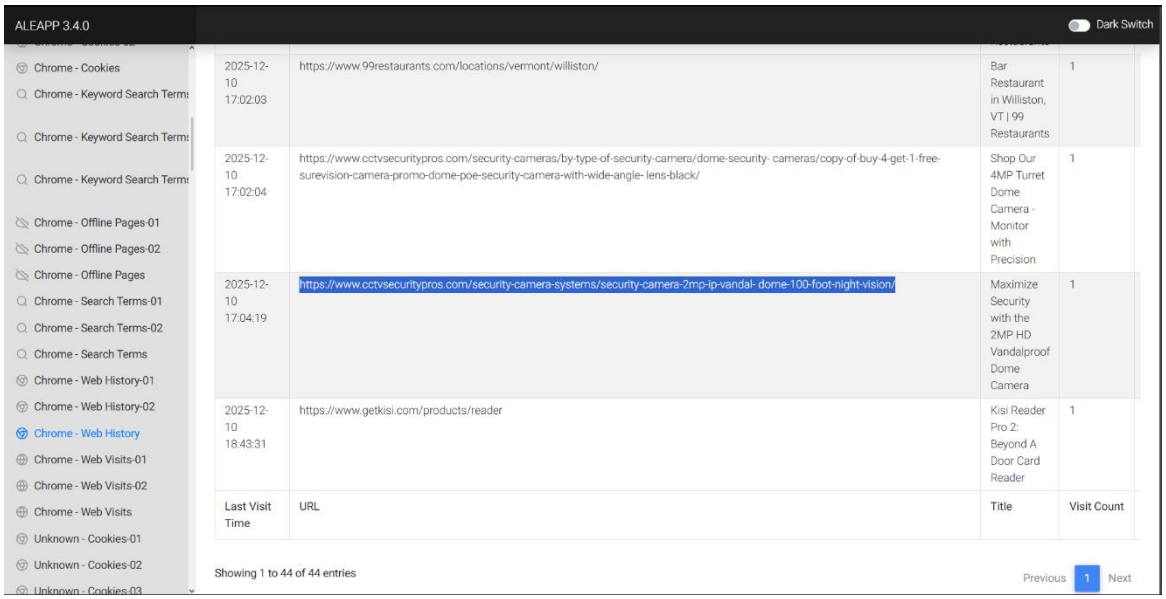


Figure x: Chrome Web History report via ALEAPP

A visited link to CCTV Security Pros was found, which led to a product page for a security camera with night vision.

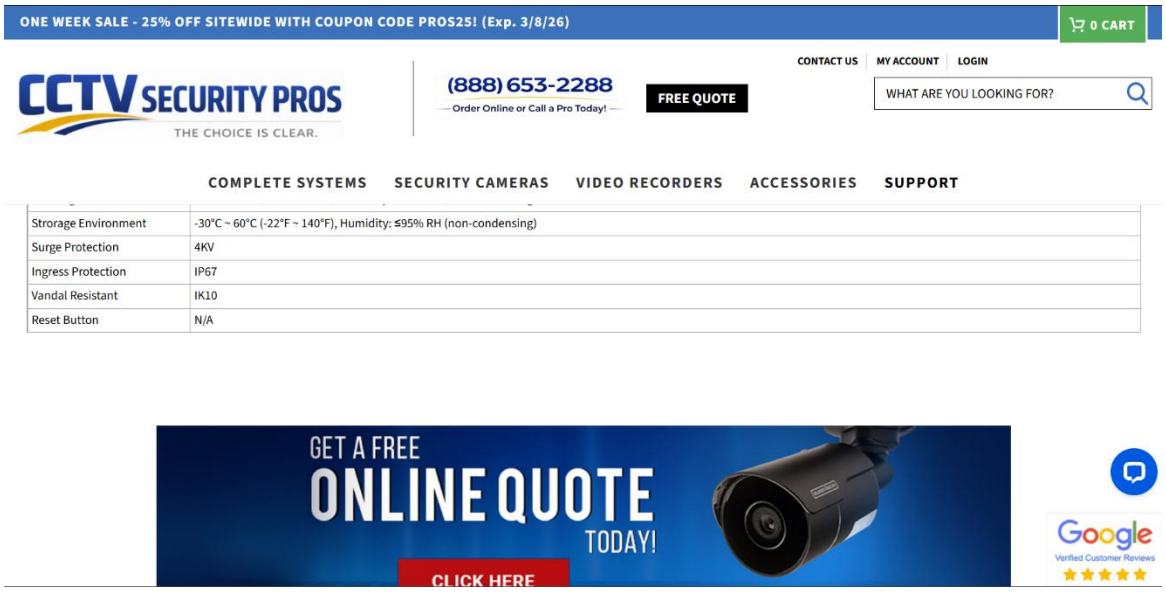


Figure x: Camera product specifications showing the flag

Upon examining the product specifications, the camera's Ingress Protection rating was identified as IP67, which was submitted as the flag.

Flag: IP67

The MVS Times (10 pts)

Question: What was the answer to the first Wordle puzzle solved?

The Crosswords database from the NYTimes application was examined using DB Browser for SQLite. In the active_puzzles table, entries with the puzzle type WORDLE were identified and sorted by date to determine the earliest puzzles played by the user.

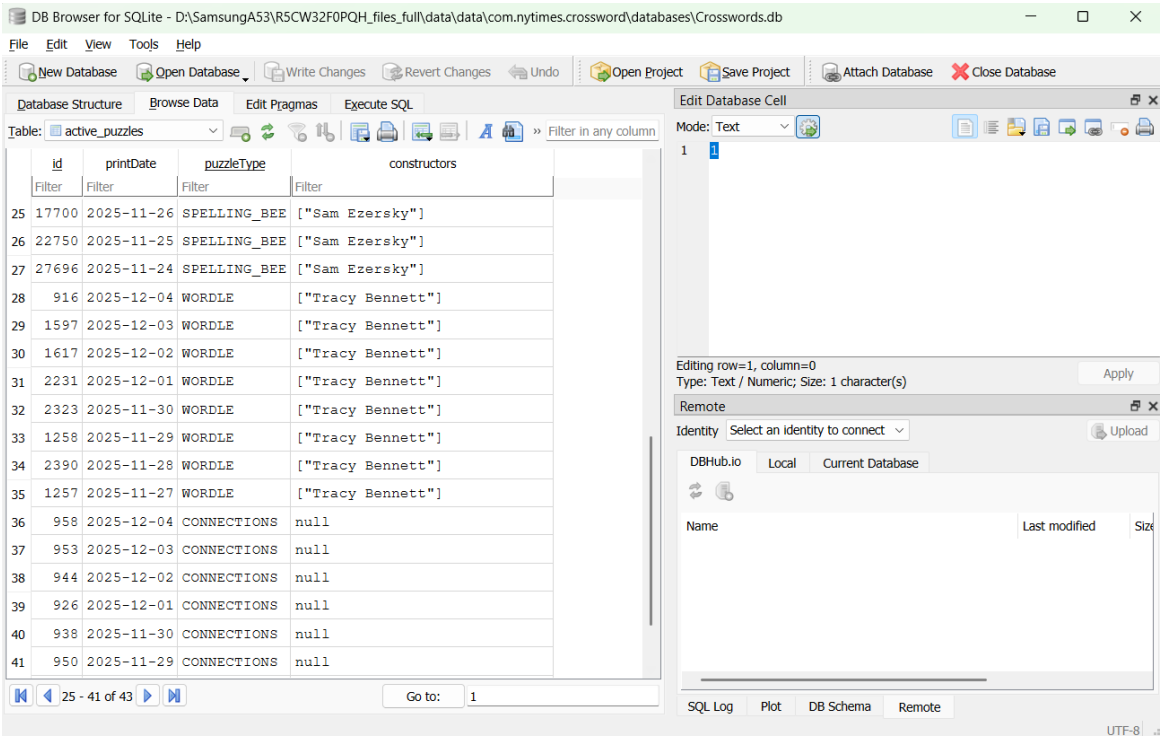


Figure x: active_puzzles table via DB Browser (SQLite)

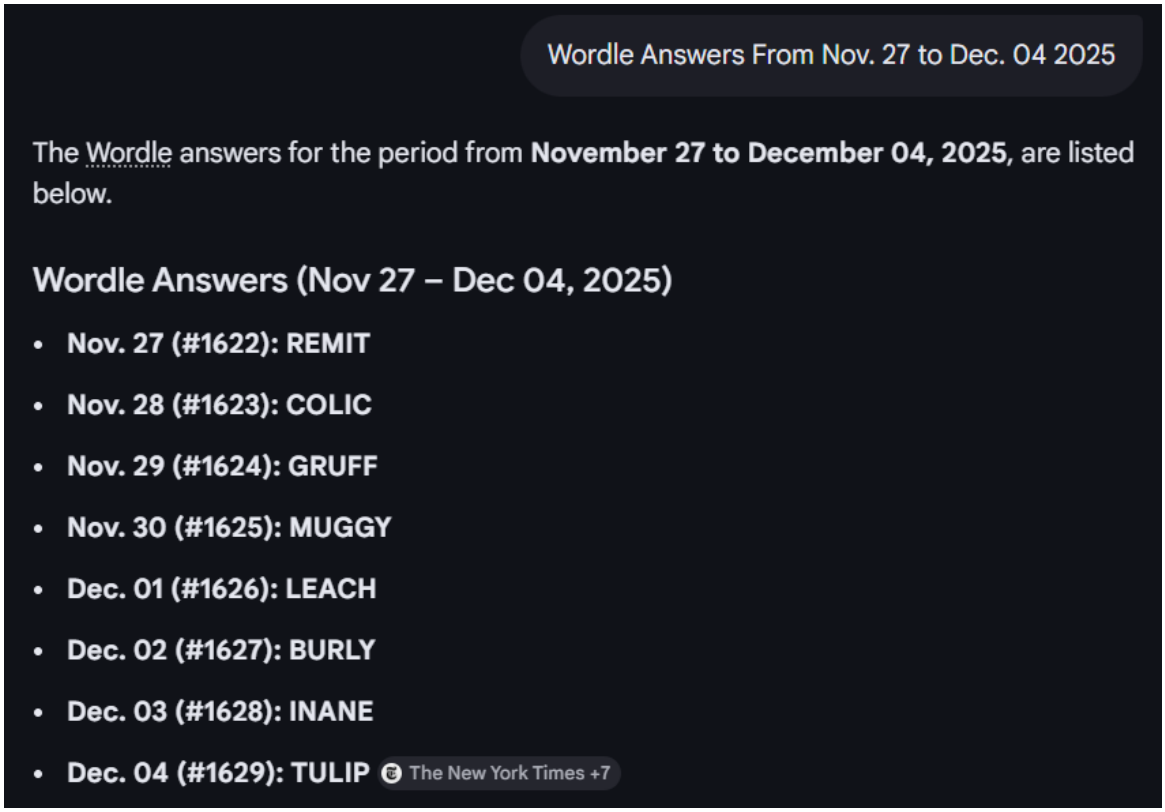


Figure x: Nov 27 – Dec 04, 2025 Wordle Answers via Google Gemini

By referencing the official Wordle answer list for that date, the solution COLIC was identified and submitted as the flag.

Flag: COLIC

Secure Attachment (10 pts)

Question: What is the name of the client the user was most recently working on?

The Samsung A53 image was analyzed using ALEAPP v3.4.0 to extract Android forensic artifacts and generate a report. In the ALEAPP report, the Gmail - App Emails section was examined to review the user's email communications.

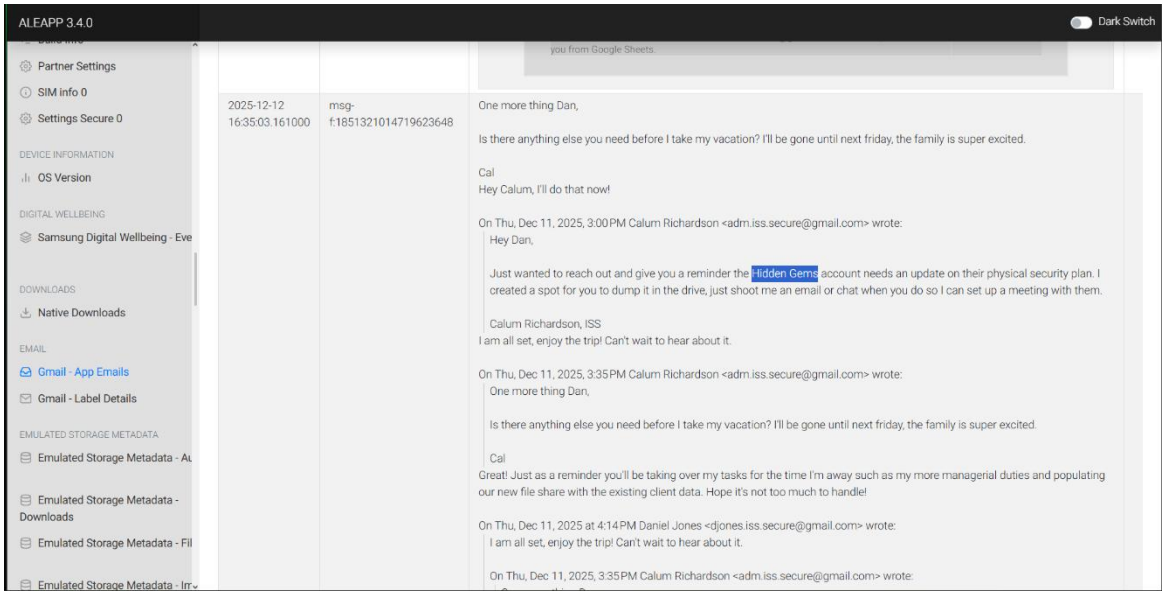


Figure x: Gmail - App Emails report via ALEAPP

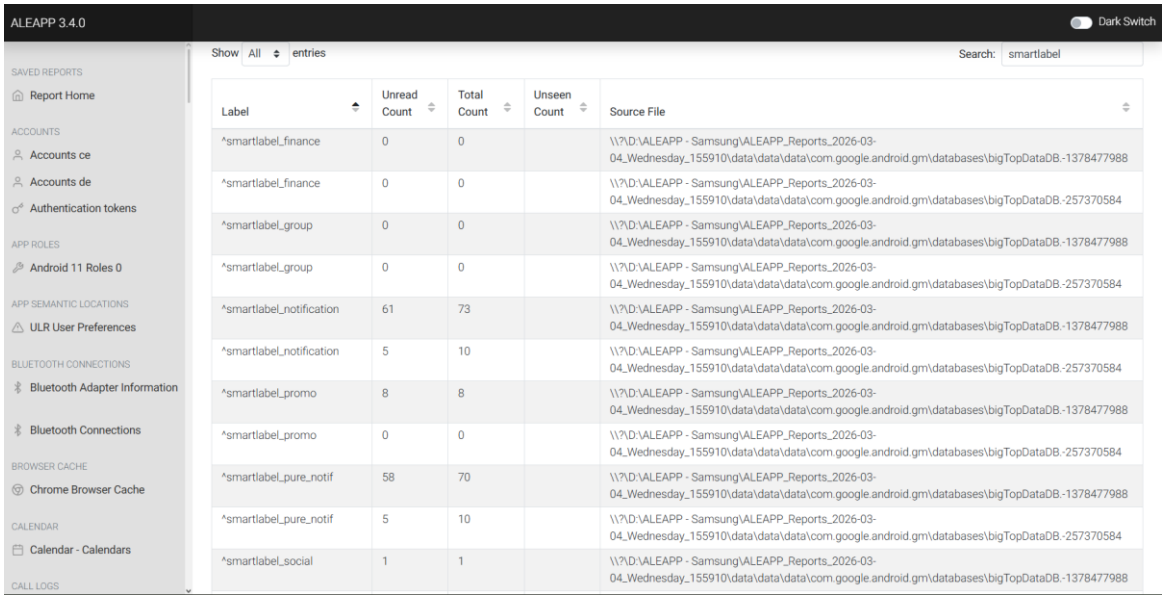
A recent email conversation referenced a client account that required an update to their physical security plan. The client name Hidden Gems was identified in the email content and submitted as the flag.

Flag: Hidden Gems

Smarty Pants (25 pts)

Question: How many emails were automatically labeled as smartlabel notifications?

The Samsung A53 image was analyzed using ALEAPP v3.4.0 to extract Android forensic artifacts and generate a report. In the ALEAPP report, the Gmail - Label Details section was examined to review the user's email labels.



Label	Unread Count	Total Count	Unseen Count	Source File
^smartlabel_finance	0	0		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-1378477988
^smartlabel_finance	0	0		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-257370584
^smartlabel_group	0	0		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-1378477988
^smartlabel_group	0	0		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-257370584
^smartlabel_notification	61	73		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-1378477988
^smartlabel_notification	5	10		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-257370584
^smartlabel_promo	8	8		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-1378477988
^smartlabel_promo	0	0		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-257370584
^smartlabel_pure_notif	58	70		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-1378477988
^smartlabel_pure_notif	5	10		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-257370584
^smartlabel_social	1	1		\\?\D:\ALEAPP - Samsung\ALEAPP_Reports_2026-03-04_Wednesday_155910\data\data\data\com.google.android.gm\databases\bigTopDataDB.-1378477988

Figure x: Gmail – Label Details report via ALEAPP

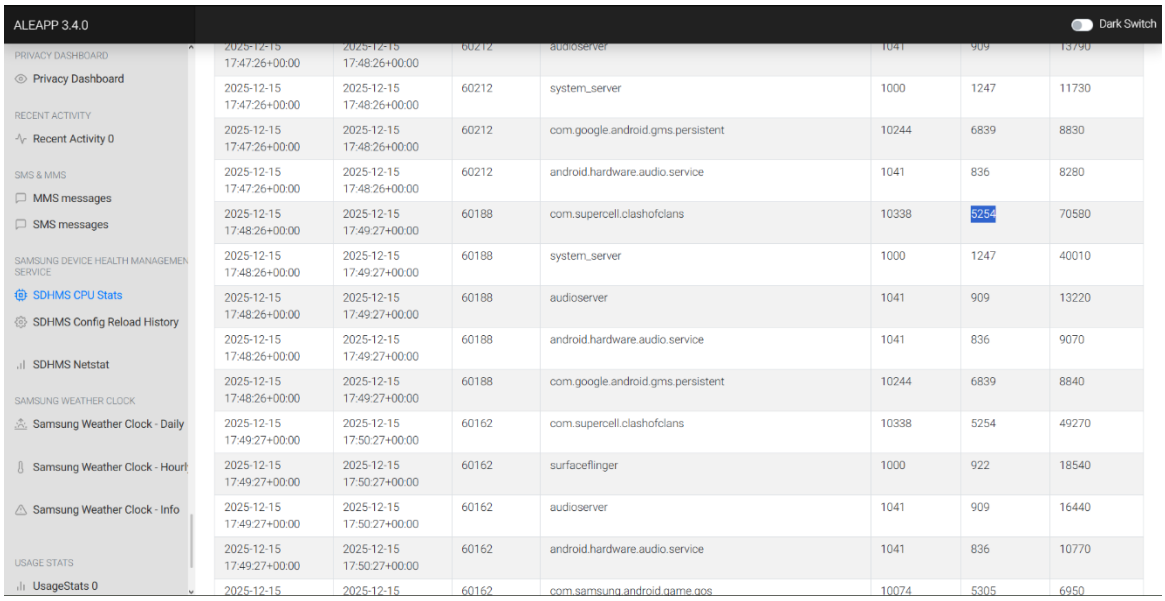
The label ^smartlabel_notification appeared twice because Gmail maintains multiple label databases. The total counts (73 and 10) were combined to determine the total number of emails automatically labeled as smartlabel notifications, resulting in 83 emails.

Flag: 83

Some Data Hides Many STATs (50 pts)

Question: What was the PID of the app that was in focus at 2025-12-15 17:48:09?

The SDHMS CPU Stats artifact in the ALEAPP report was examined to identify active processes around 2025-12-15 17:48:09 (UTC).



2025-12-15 17:47:26+00:00	2025-12-15 17:48:26+00:00	60212	audioserver	1041	909	13790
2025-12-15 17:47:26+00:00	2025-12-15 17:48:26+00:00	60212	system_server	1000	1247	11730
2025-12-15 17:47:26+00:00	2025-12-15 17:48:26+00:00	60212	com.google.android.gms.persistent	10244	6839	8830
2025-12-15 17:47:26+00:00	2025-12-15 17:48:26+00:00	60212	android.hardware.audio.service	1041	836	8280
2025-12-15 17:48:26+00:00	2025-12-15 17:49:27+00:00	60188	com.supercell.clashofclans	10338	5254	70580
2025-12-15 17:48:26+00:00	2025-12-15 17:49:27+00:00	60188	system_server	1000	1247	40010
2025-12-15 17:48:26+00:00	2025-12-15 17:49:27+00:00	60188	audioserver	1041	909	13220
2025-12-15 17:48:26+00:00	2025-12-15 17:49:27+00:00	60188	android.hardware.audio.service	1041	836	9070
2025-12-15 17:48:26+00:00	2025-12-15 17:49:27+00:00	60188	com.google.android.gms.persistent	10244	6839	8840
2025-12-15 17:49:27+00:00	2025-12-15 17:50:27+00:00	60162	com.supercell.clashofclans	10338	5254	49270
2025-12-15 17:49:27+00:00	2025-12-15 17:50:27+00:00	60162	surfaceflinger	1000	922	18540
2025-12-15 17:49:27+00:00	2025-12-15 17:50:27+00:00	60162	audioserver	1041	909	16440
2025-12-15 17:49:27+00:00	2025-12-15 17:50:27+00:00	60162	android.hardware.audio.service	1041	836	10770
2025-12-15	2025-12-15	60162	com.samsung.android.game.gos	10074	5305	6950

Figure x: SDHMS CPU Stats report via ALEAPP

The process com.supercell.clashofclans was running between 17:48:26 and 17:49:27, indicating it was the app in focus during that timeframe. The corresponding PID shown in the table was 5254, which is the flag.

Flag: 5254

Out of Bounds (70 pts)

Question: What was the top package name in a splitscreen shortcut?

The file apps_edge_sa_status_pref.xml was examined, which stores Samsung Edge Panel split-screen shortcut configurations. The entry EG_2033 contained paired application components indicating split-screen app pairs. This was verified in DB Browser for SQLite by examining the appsedge.db database, where the component_name column showed entries such as com.google.android.apps.messaging/... and com.google.android.youtube/.... From the first split-screen pair com.google.android.apps.messaging/com.google.android.youtube, the top package name identified was com.google.android.youtube

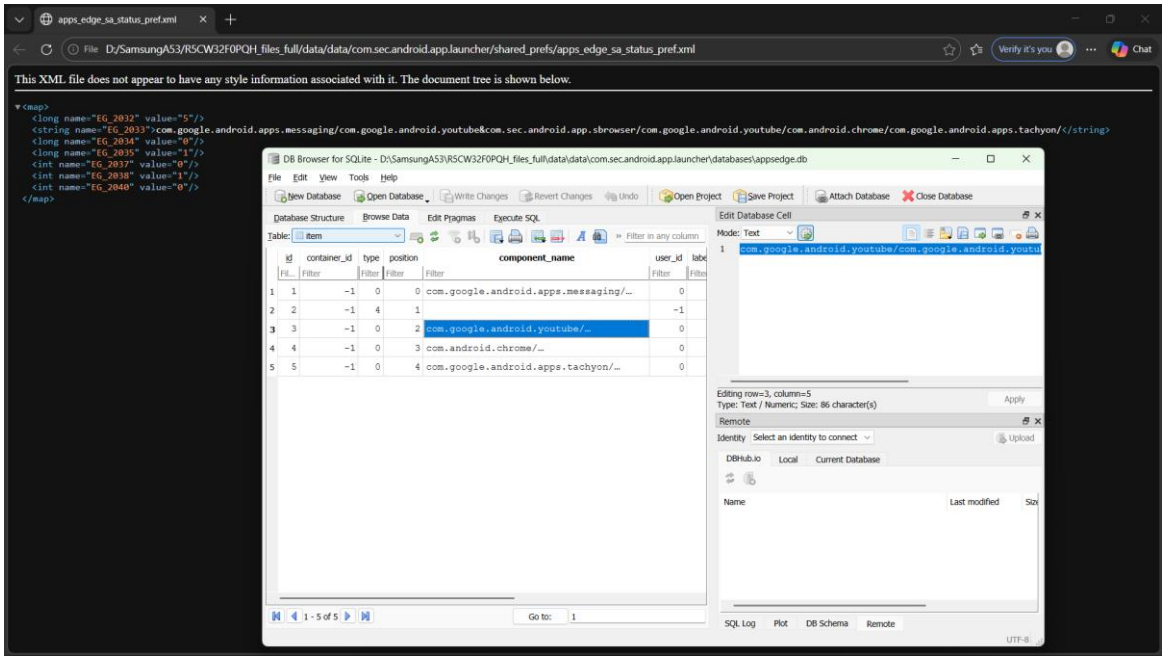


Figure x: apps_edge_sa_status_pref.xml content and item table in appsedge.db via DB Browser (SQLite)

Flag: com.google.android.youtube

IOS Challenges

Need a Challenge? (5 pts)

Question: What Instagram handle does User ID 2278169415 belong to?

The hint for this question is the User ID itself, with the use of Google Gemini, it was able to figure out whose account is assigned to that unique user ID.

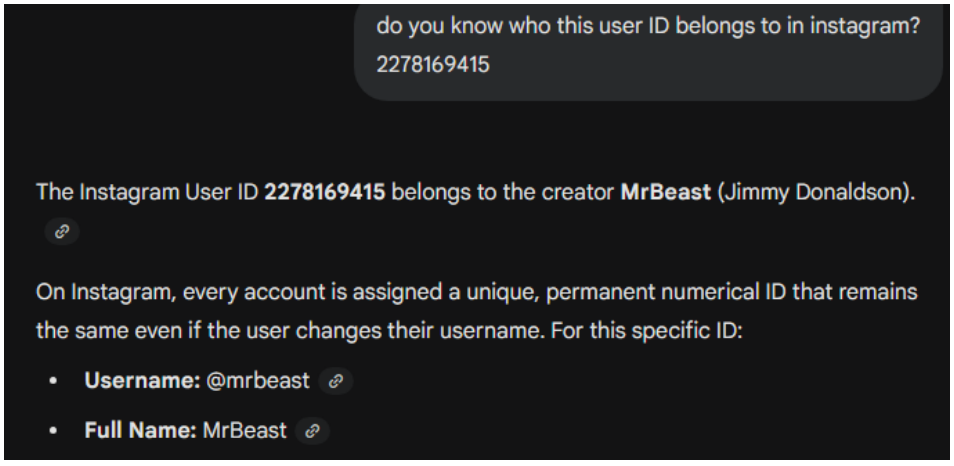


Figure x: AI Prompt to the Instagram User ID

Flag: mrbeast

So Thoughtful (5 pts)

Question: What item does the user need to buy for Christmas?

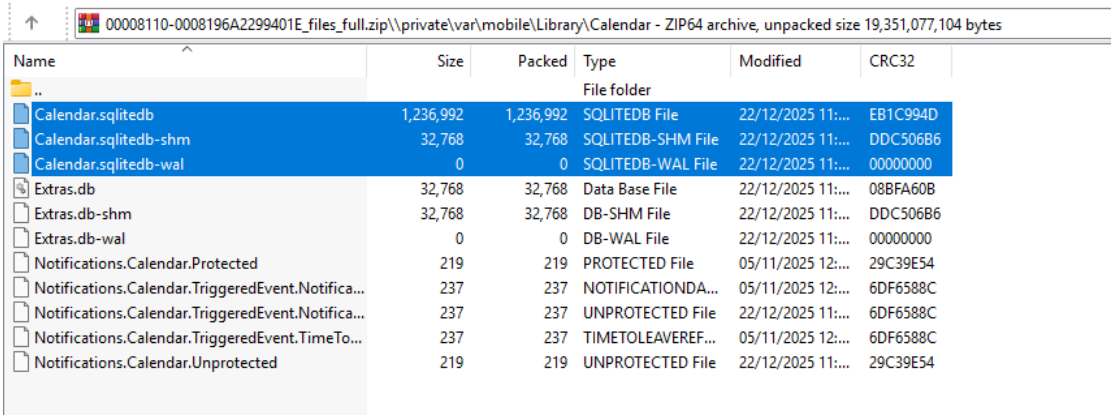


Figure x: Directory to the Calendar folder along with the database file

In this case, we would assume that the item would probably be in the Reminders app. So, the first database to look at is the Reminders database by looking at the directory to:

```
/private/var/mobile/Library/Calendar/
```

Look for **Calendar.sqlitedb**. Open it with SQLite and query as to where the item would be. Based on the database structure, we could look for a keyword to specify the holiday since the user would need to buy something for Christmas.

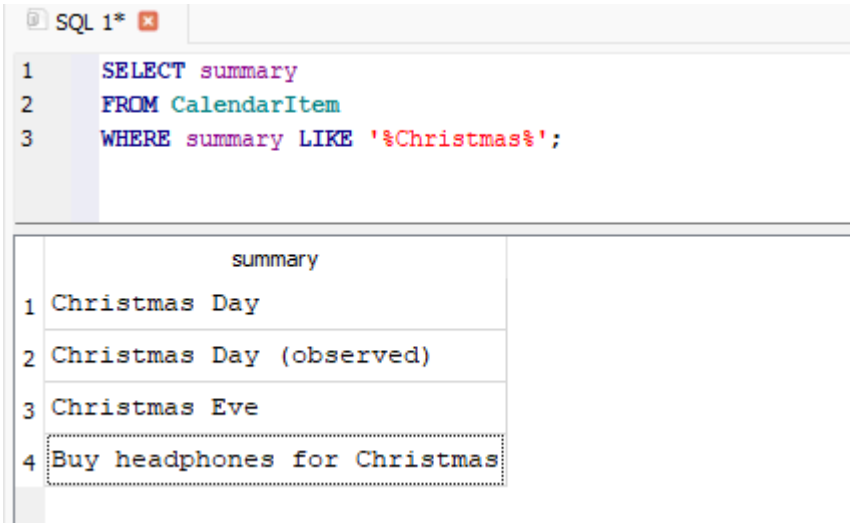


Figure x: Query from the Calendar.sqlitedb

Flag: headphones

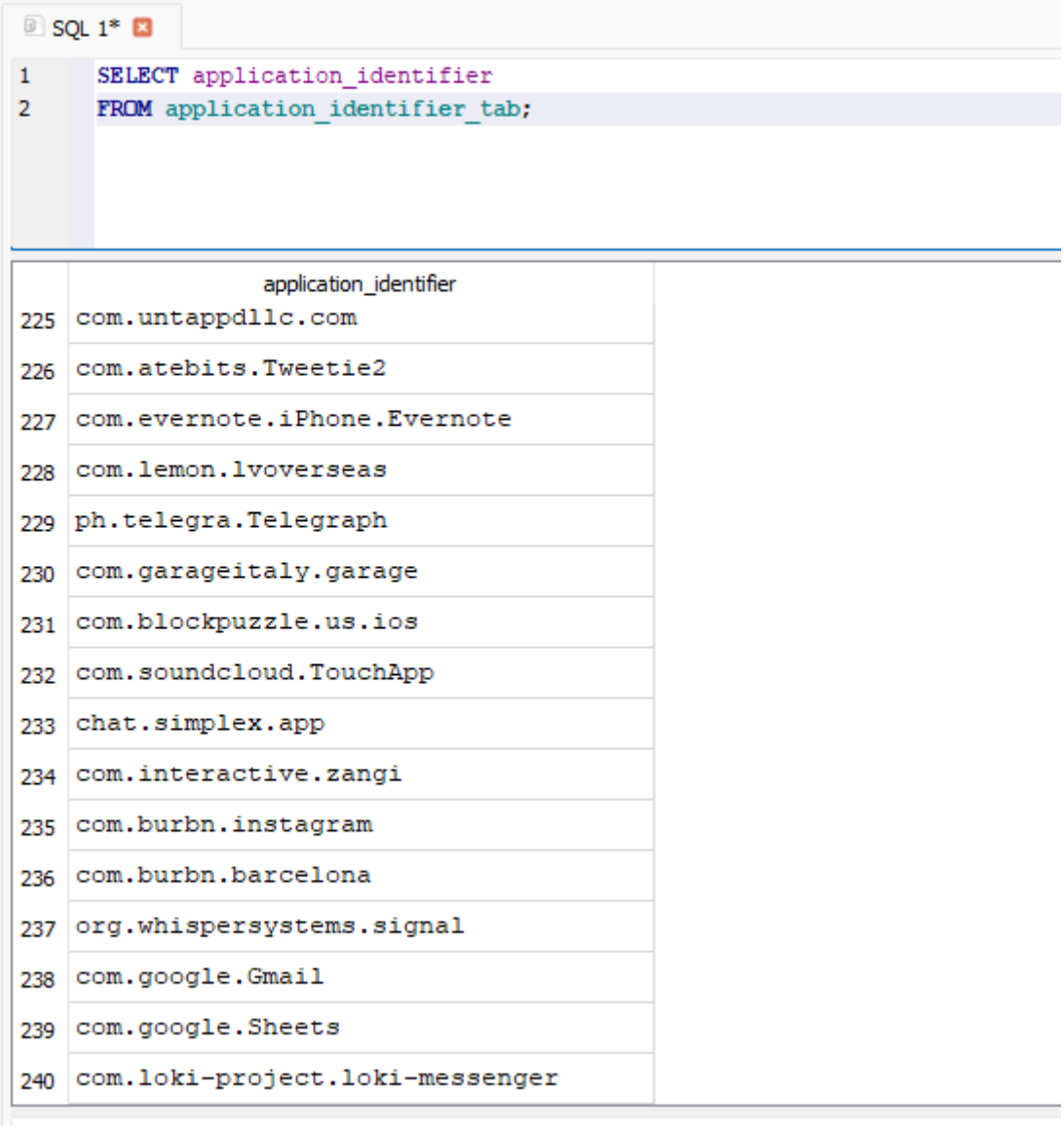
Do you have any games on your phone? (10 pts)

Question: What genre of app did the user download the most of?

In a newer version of IOS (16/17+), all installed apps are recorded in the applicationState.db. According to Gemini, this database tracks all installed apps. In this case, we opened up winrar to look for a location:

`/private/var/mobile/Library/FrontBoard/`

Then, by following up with a query by selecting application_identifier in the table, it shows up a list of apps.



SQL 1*

```
1  SELECT application_identifier
2  FROM application_identifier_tab;
```

	application_identifier
225	com.untappdllc.com
226	com.atebits.Tweetie2
227	com.evernote.iPhone.Evernote
228	com.lemon.lvoverseas
229	ph.telegra.Telegraph
230	com.garageitaly.garage
231	com.blockpuzzle.us.ios
232	com.soundcloud.TouchApp
233	chat.simplex.app
234	com.interactive.zangi
235	com.burbn.instagram
236	com.burbn.barcelona
237	org.whispersystems.signal
238	com.google.Gmail
239	com.google.Sheets
240	com.loki-project.loki-messenger

Figure x: Query from the Cache database

We sent the list of apps in ChatGPT to classify app genres and the most one would be Social Networking.

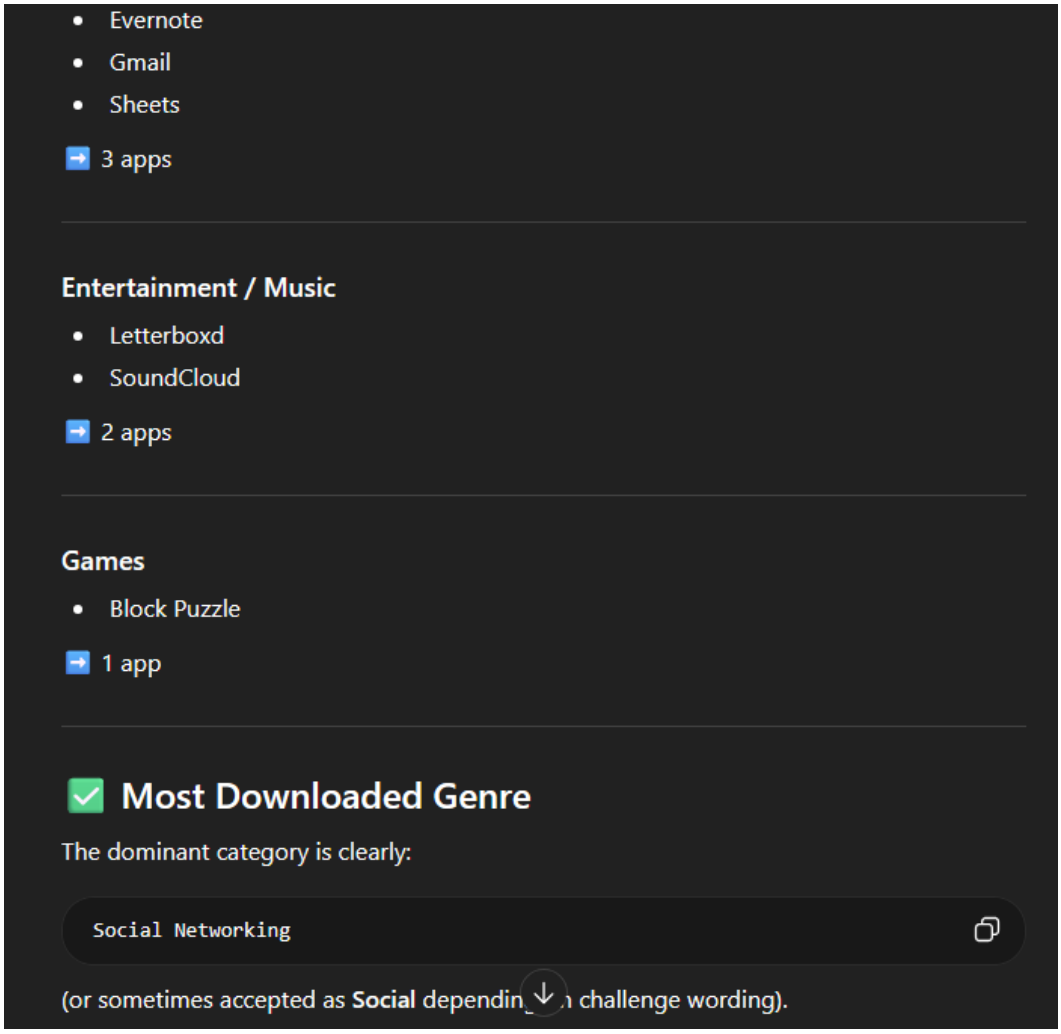


Figure x: Classifying apps from ChatGPT

Flag: Social Networking

That would look nice in my garage (10 pts)

Question: What is the make and model of the car that was photographed without a iPhone?

Because a photograph was mentioned, we searched the Media folder inside the IOS zipped file. Inside the Media folder, we just looked for every sub-folder that was inside and found an image of a car embedded in a series of folders.

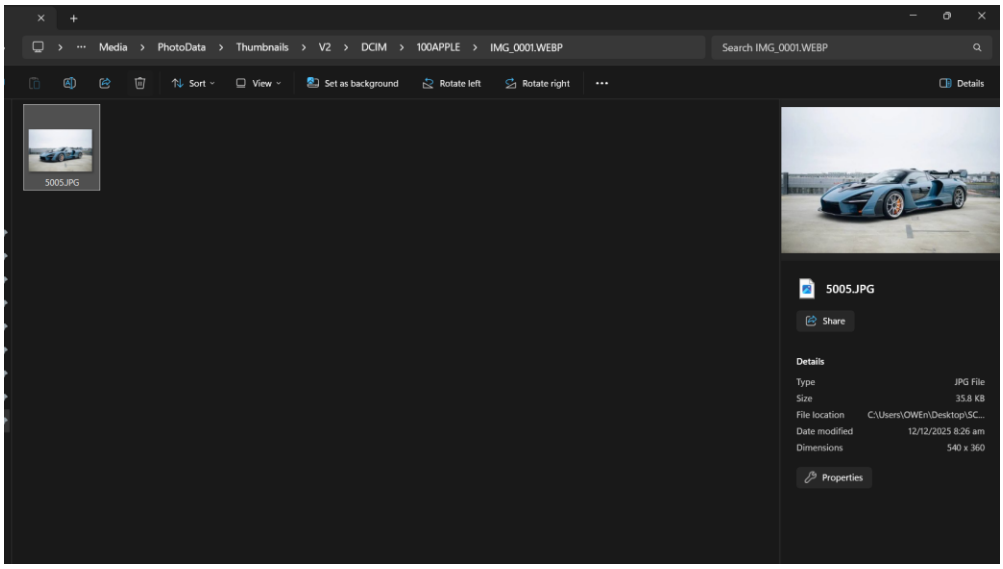


Figure x: Screenshot of location of the car

Using help from Google Gemini, the make and model of the car was found.

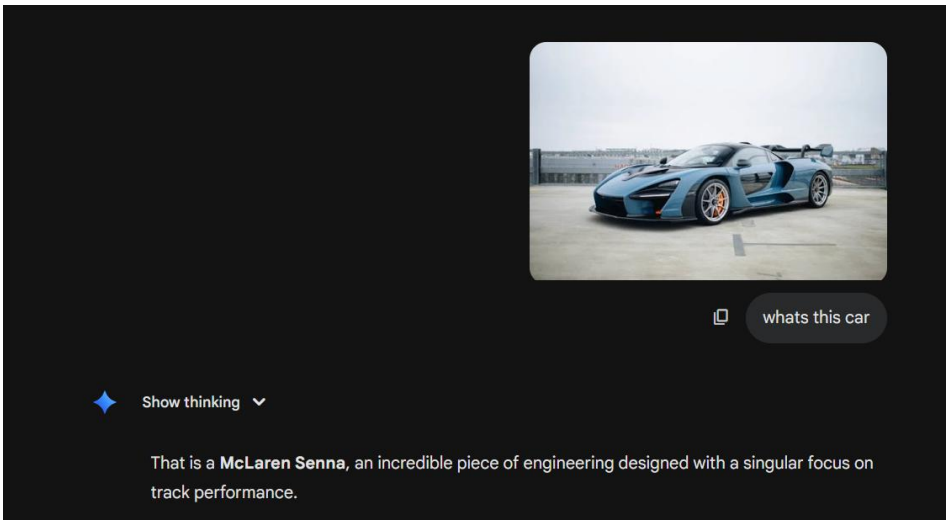


Figure x: Google Gemini car prompt

Flag: McLaren Senna

Where are we going? (10 pts)

Question: What airport was the phone near when it pinged off of a cell tower on 2025-12-16 22:47:24 (UTC)?

In iOS forensics, cell tower "pings" are typically recorded in a cache database managed by the location of daemon. So, the directory would be in `\private\var\mobile\Library\Caches\com.apple.routined\` giving a Cache.sqlite file.

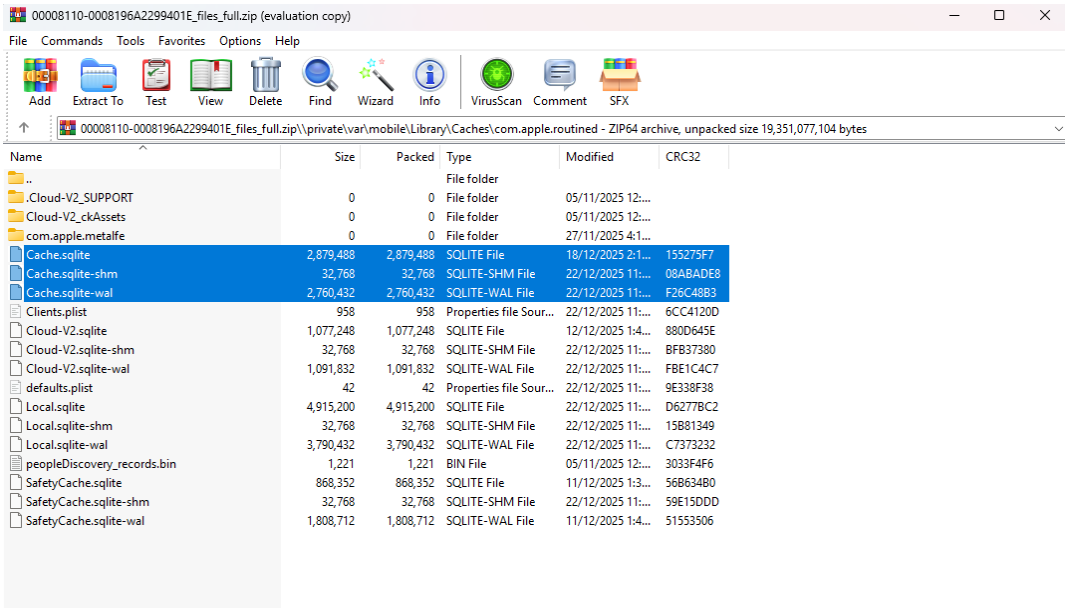
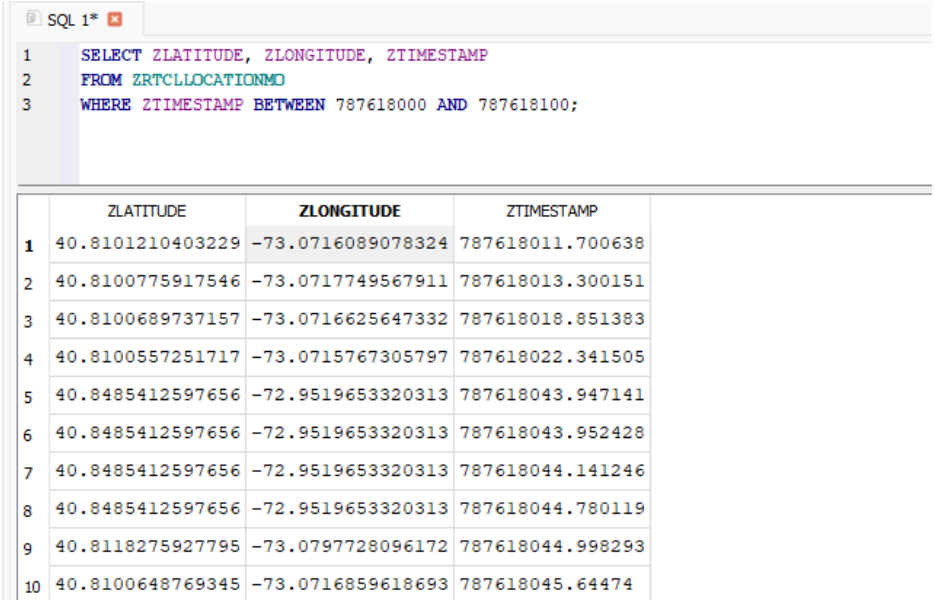


Figure x: Cache directory for location

Write a query from Gemini to locate the coordinates within that time and provide a full list of coordinates from that time.



	ZLATITUDE	ZLONGITUDE	ZTIMESTAMP
1	40.8101210403229	-73.0716089078324	787618011.700638
2	40.8100775917546	-73.0717749567911	787618013.300151
3	40.8100689737157	-73.0716625647332	787618018.851383
4	40.8100557251717	-73.0715767305797	787618022.341505
5	40.8485412597656	-72.9519653320313	787618043.947141
6	40.8485412597656	-72.9519653320313	787618043.952428
7	40.8485412597656	-72.9519653320313	787618044.141246
8	40.8485412597656	-72.9519653320313	787618044.780119
9	40.8118275927795	-73.0797728096172	787618044.998293
10	40.8100648769345	-73.0716859618693	787618045.64474

Figure x: Query from Gemini to get the coordinates

We put these coordinates in Google Maps, and the ping was located near the Long Islang MacArthur Airport.

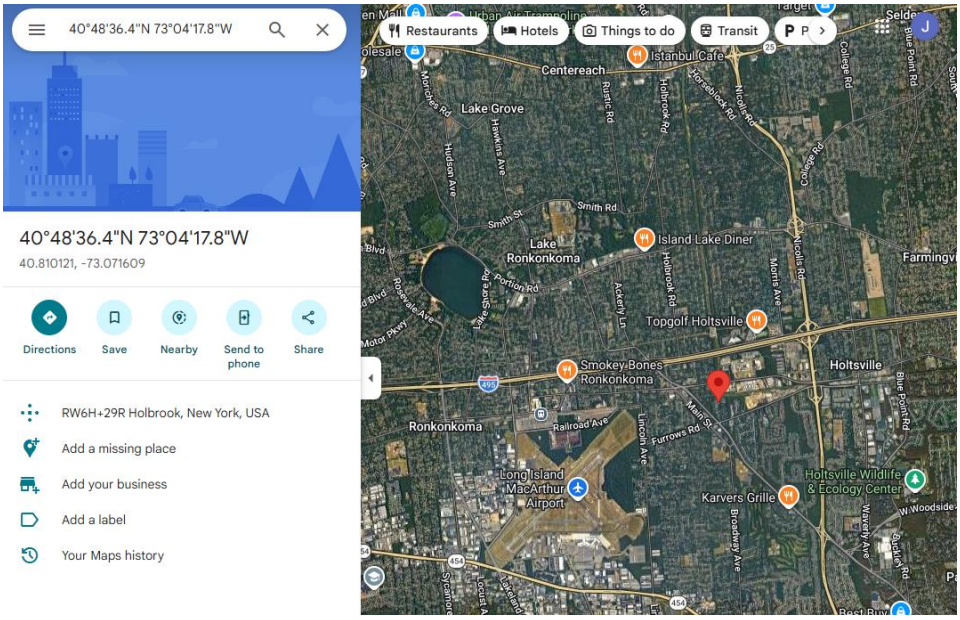


Figure x: Ping location of the coordinates given from the query

Flag: Long Island MacArthur Airport

Cipher Challenges

Can You Handle This One (10 pts)

Given:

Hi Jessica! 00110110 00111000 00100000 00110111 00110100 00100000 00110111 00110100 00100000 00110111 00110000 00100000 00110111 00110011 00100000 00110011 01100001 00100000 00110010 01100110 00100000 00110010 01100110 00100000 00110111 00110011 00100000 00110111 00110100 00100000 00110110 00110001 00100000 00110111 00110010 00100000 00110111 00110100 00100000 00110110 01100100 00100000 00110110 00110101 00100000 00110010 01100101 00100000 00110111 00110011 00100000 00110111 00110100 00100000 00110110 00110001 00100000 00110111 00110010 00100000 00110110 01100010 00100000 00110011 00110100 00100000 00110110 01100101 00100000 00110011 00110110 00100000 00110010 01100101 00100000 00110110 00110011 00100000 00110110 01100110 00100000 00110110 01100100 00100000 00110010 01100110

To solve this problem, I asked for the help of ChatGPT to decode the binary as ASCII.

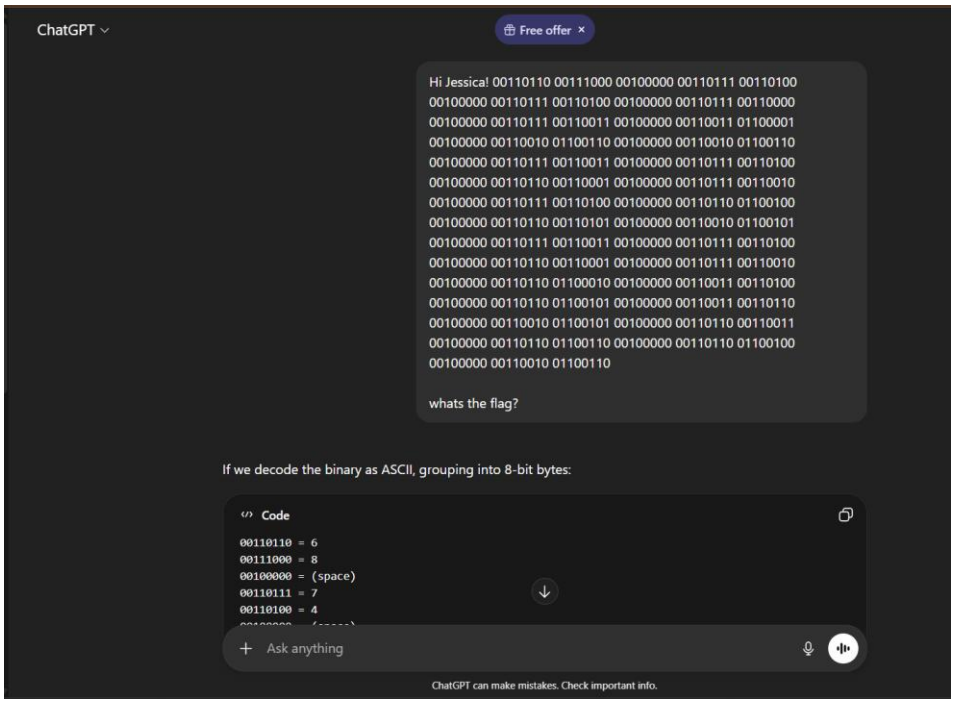


Figure x: Screenshot of the Prompt with ChatGPT

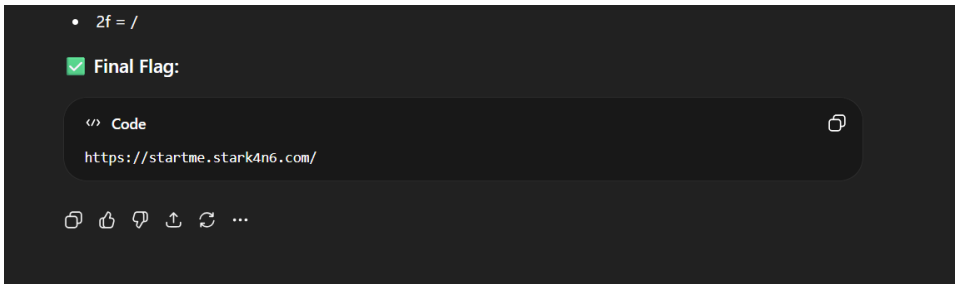


Figure x: Screenshot of ChatGPT's response

Flag: https://startme.stark4n6.com/

Peel Back the Layers (25 pts)

Question: Can you find the hidden link? (Provide the redirected resource URL)

Given: ThisIsFine.pxz

To solve this, Pixlr, an online image converter was used to convert the pxz file into an openable png format.



Figure x: The image when converted into png

Inside the image, below the thumbs up, a link was embedded, which would be: http://tiny.cc/nh5v001. Since the question was asking for the redirected resource URL, the link was opened which was a redirect to another website: https://github.com/abrignon.

Flag: https://github.com/abrignon

Good Grief! (25 pts)

Question: The Great Pumpkin's Inverse Logic

Given: VanPelt.png



Figure x: VanPelt.png

To solve this, a Pigpen Cipher key was used. However, since the clue mentions inversion, I figured out that I would need to either inverse the orderings of the cipher, or I would inverse the orderings of the key.

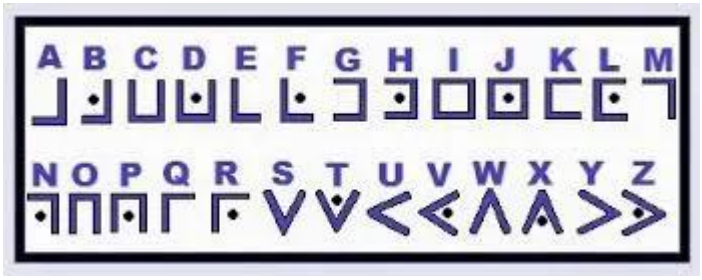


Figure x: Image of the Pigpen Cipher key used

I thought that inversing the orderings of the key was more probable as it would be a more fitting considering the question was at 25 points. Inversing only the orderings of the Cipher would be too simple for a question that had that many points. This would mean that what cipher was supposed to represent A would now represent Z, the key that represents B would now represent Y, what represented C would represent X, and so on. Because creating the inverted key would consume too much time, I manually checked each cipher key for its new representation and got a readable word.

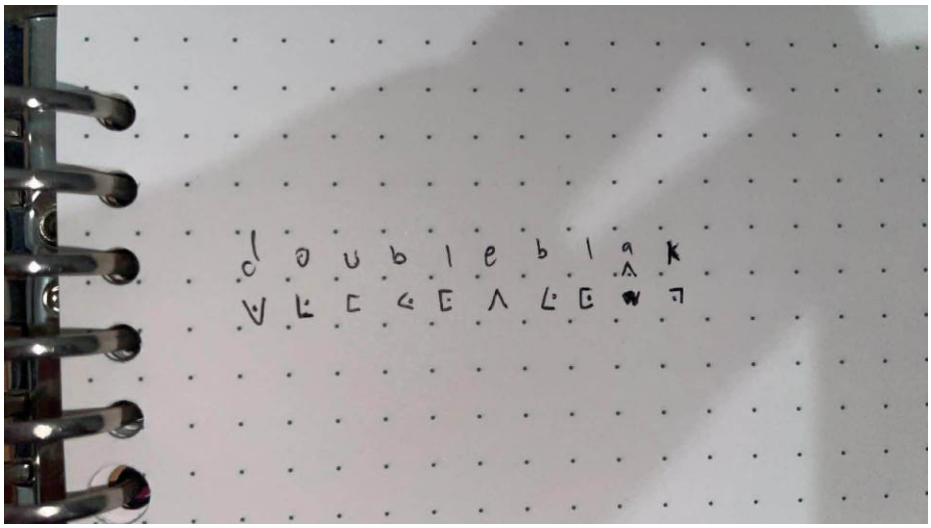


Figure x: The cipher decoded using the inverted key

Flag: doubleblak

Resident Evil (50 pts)

Question: Can you find the link?

Given: Challenge3.vhdk

Because this is a virtual disk, I opened it, and there was nothing inside except a readme.txt file. Inside the readme was a coded message.

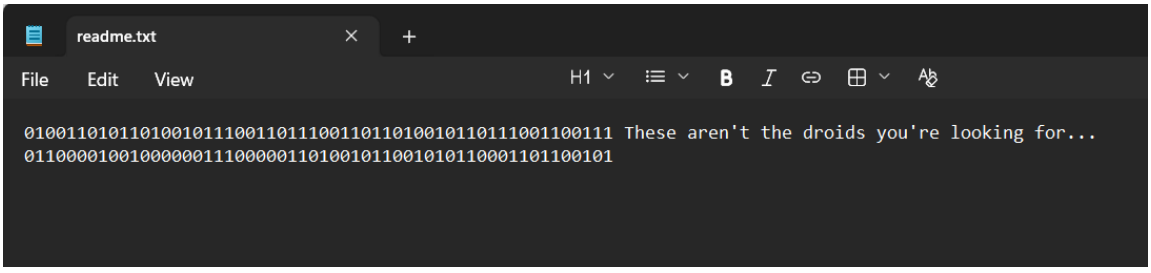


Figure x: Contents of the readme file

Translating the coded message through Google Gemini spelled out Missing a piece, which meant that the answer was not in this file, which was further reinforced by the Star Wars quote.

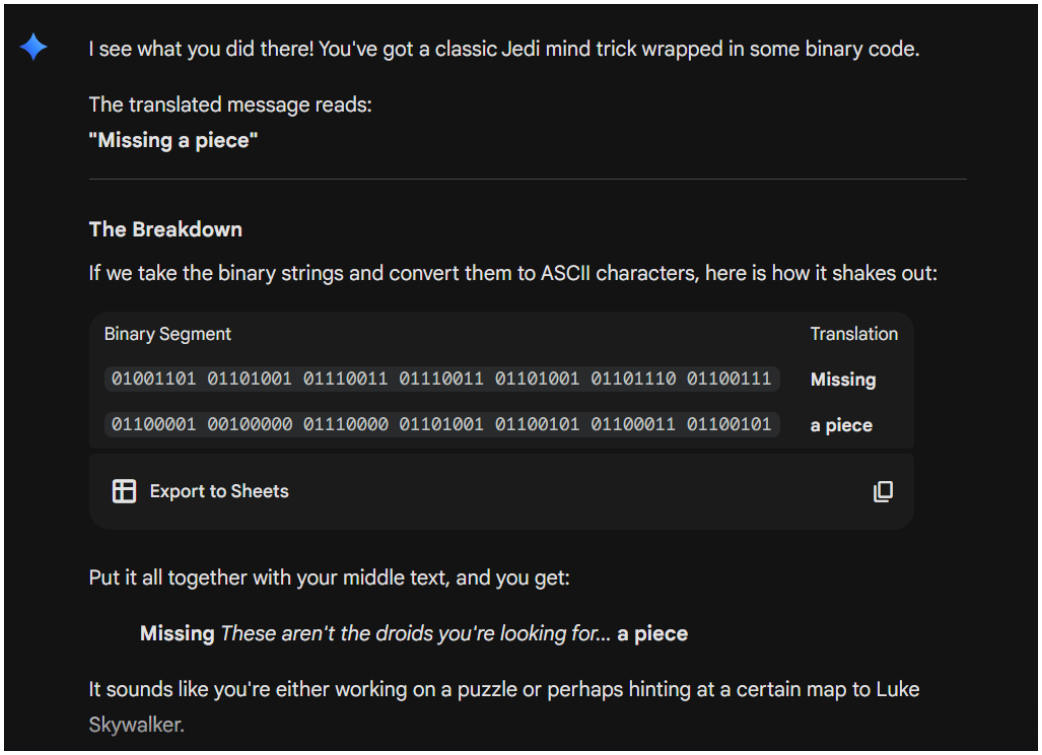


Figure x: readme.txt Google Gemini prompt

Since the given resource was a drive, I figured it was possible to mount it using FTK Imager in hopes that the answer was hidden in the drive.

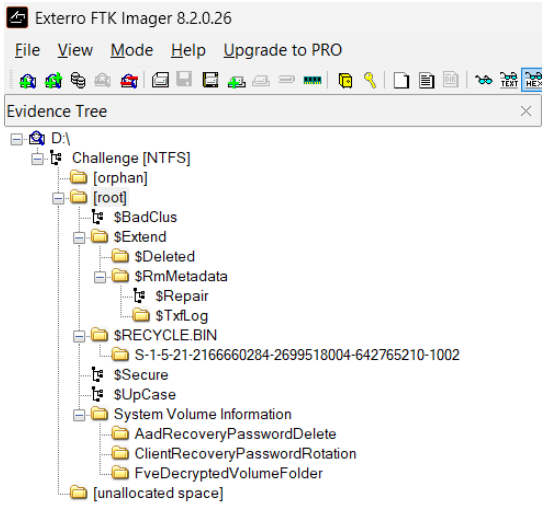


Figure x: Challenge3.vhdk added as an evidence item

After checking each file of the drive in FTK Imager, the $MFT file contained useful information. Inside the $MFT file, more binary code was seen, most of which were not seen inside the readme.txt file.

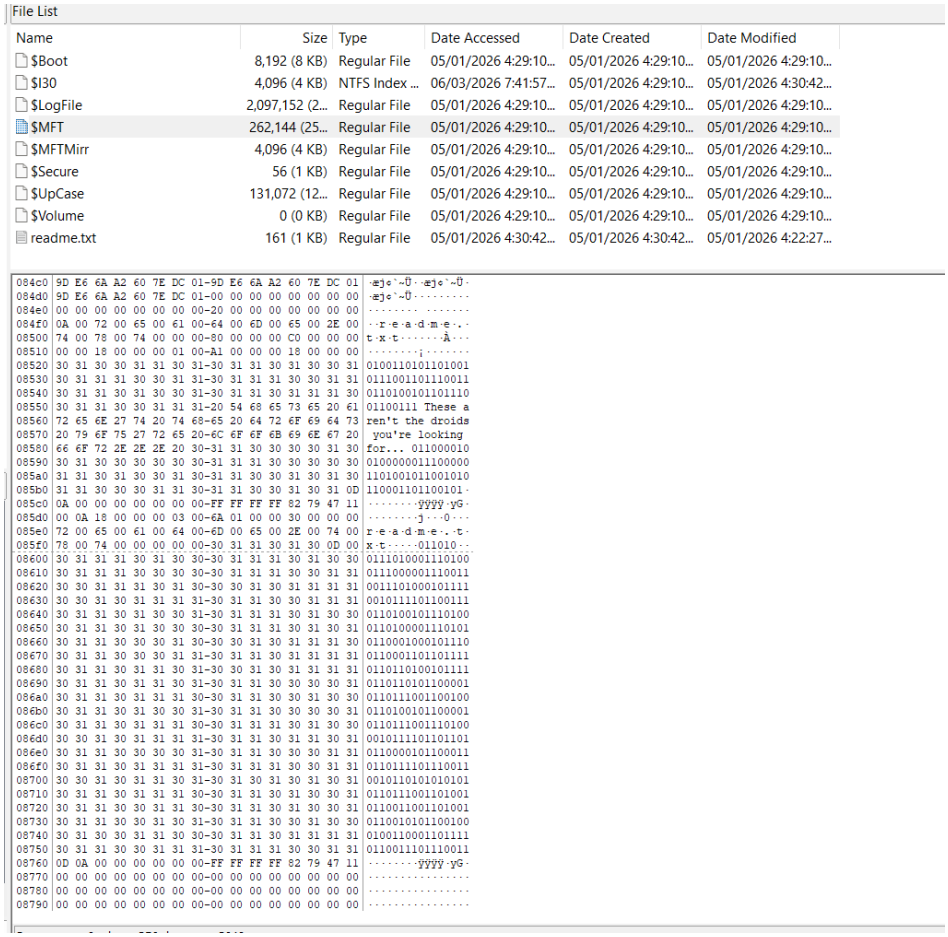
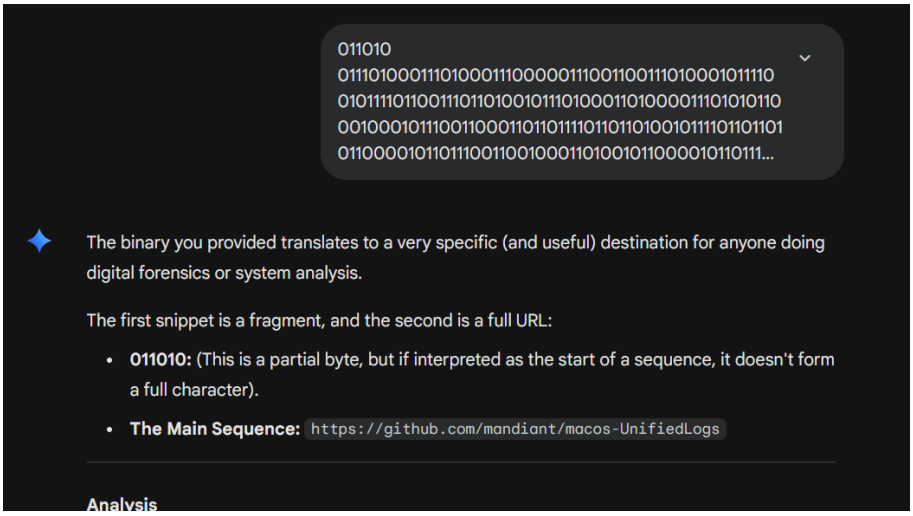


Figure x: Content of $MFT File.

Taking the newly seen binary code for inspection with the help of Google Gemini, a link was found.


011010
01110100011101000111000001110011001110100010111100101111001110110100101110100001110100101101100100001011100011001100011101101110111011010010111010000111010101100100001011100011001100011101101110111011010010111010000111010101100100001011100011001100011101101110111011010010111010000111010

Bonsoir Elliot (50 pts)

Question: It's not that Deep, just Sound it out (shout out to int80)

Given: HideNSeek.wav

Listening to the wav file did not give anything meaningful as it was just a song. Therefore, the question must have been hiding a clue. Looking at the question, only Deep and Sound were capitalized. After looking through an online forensics toolkit by stark4n6, wherein the online toolkit was also the answer to the Cipher, '*Can you Handle this one?',* a tool called DeepSound was found. After downloading the application, the file HideNSeek.wav was opened and inside, there was an embedded file.

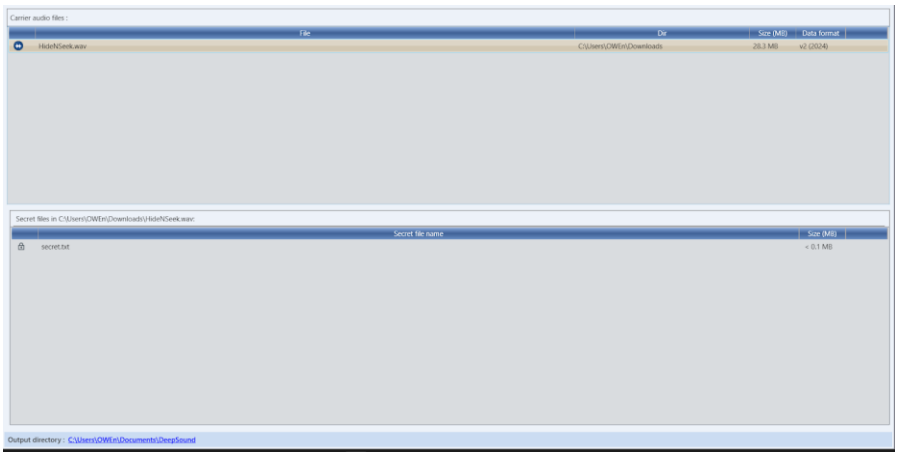


Figure x: HideNSeek.wav opened inside DeepSound

The file was named secret.txt file wherein the answer to the question was embedded inside.



Figure x: Contents of secret.txt

Flag: https://dfir.blog/unfurl/

Cows come in all shapes and sizes (75 pts)

Question: Moooo! Get the cow to spill the secret! (zip password: moo)

Given: moo.zip

The provided file for this challenge was moo.zip. After downloading the archive, it was extracted, revealing a runnable executable file named moo.exe.

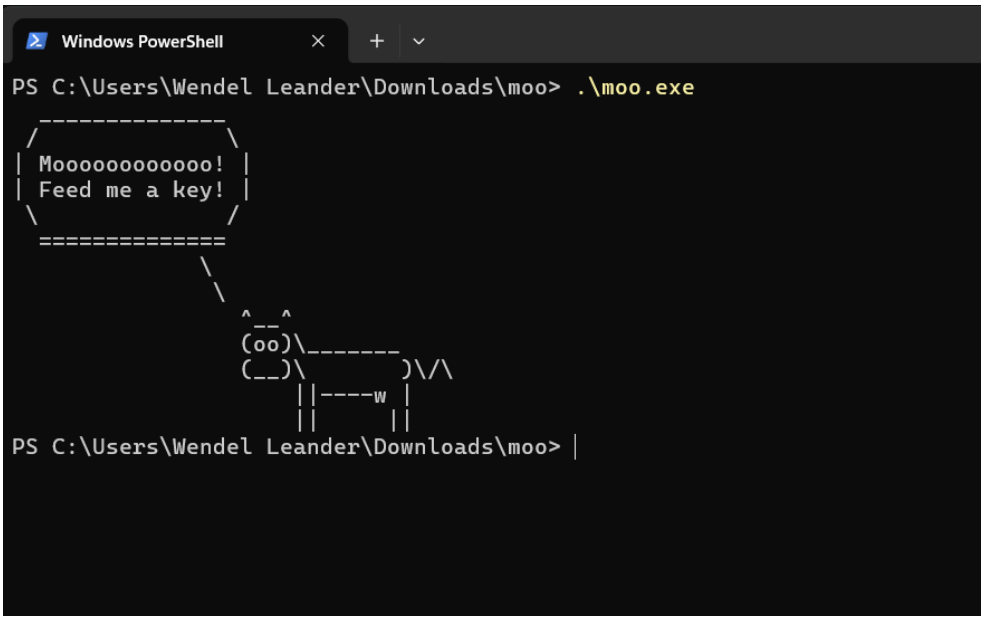


Figure x: Running moo.exe

When the executable was launched, it displayed an ASCII art image of a cow. A dialogue box appeared above the cow with a prompt instructing the user to “feed me a key.”

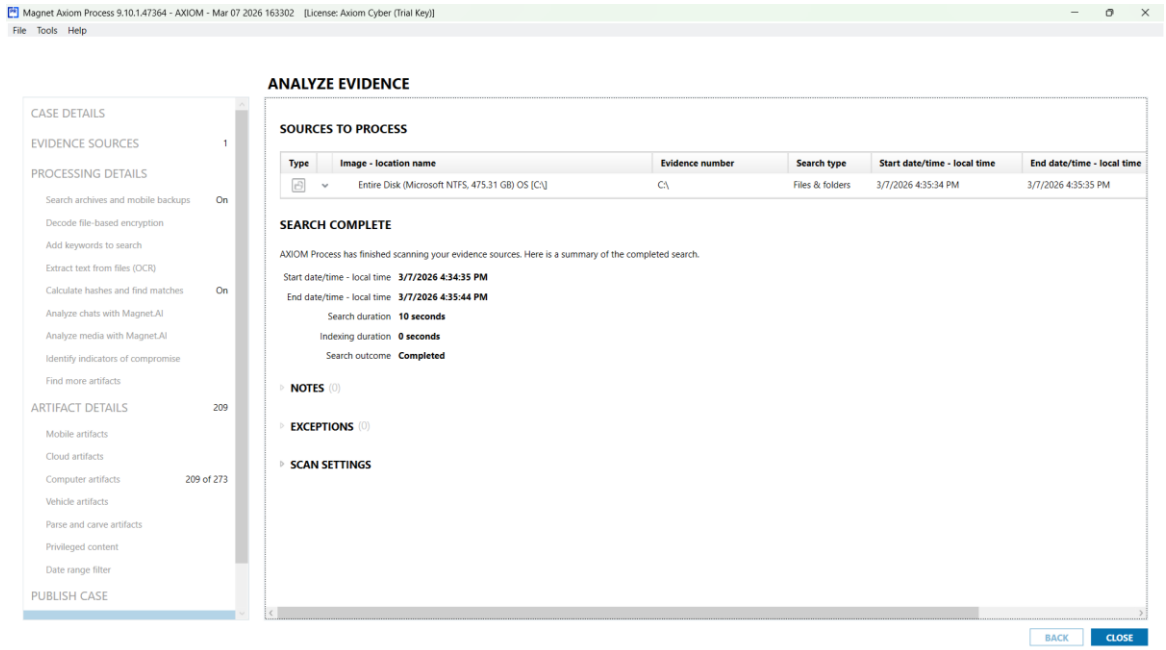


Figure x: Processing moo.exe via Magnet Axiom

To further analyze the executable, the file was processed using Magnet Axiom, a digital forensic analysis tool capable of extracting artifacts and embedded data from files.

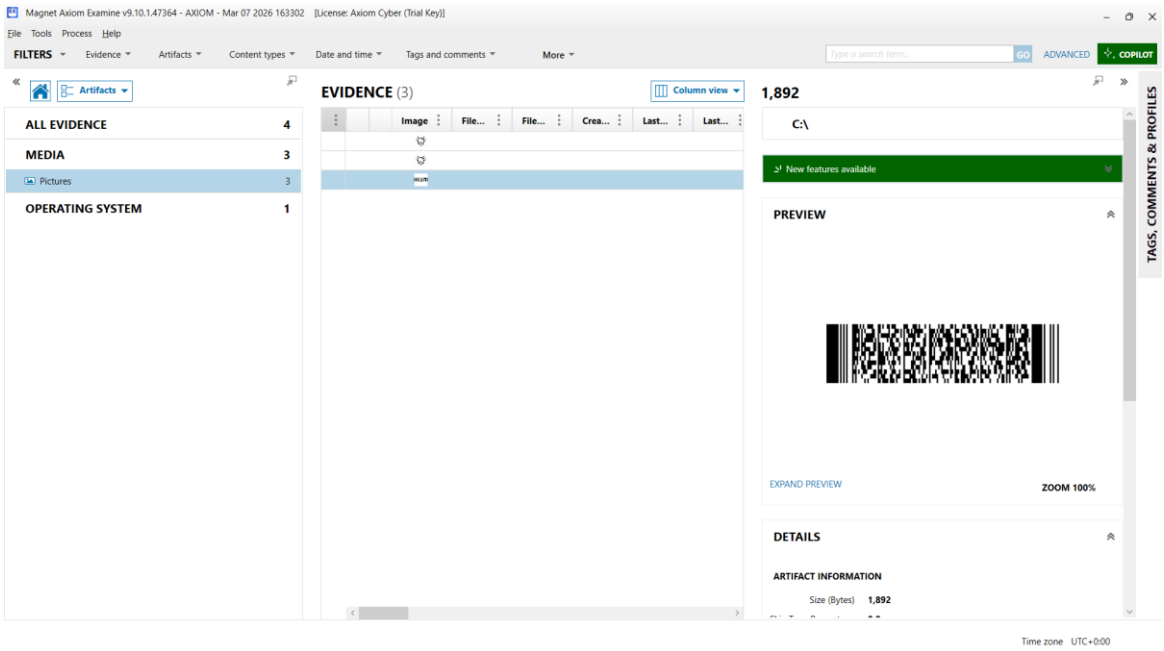


Figure x: Media evidence from moo.exe revealing bar code

After processing the executable in Magnet Axiom, several embedded graphics were identified. These graphics included images of cows in various sizes and a barcode image.

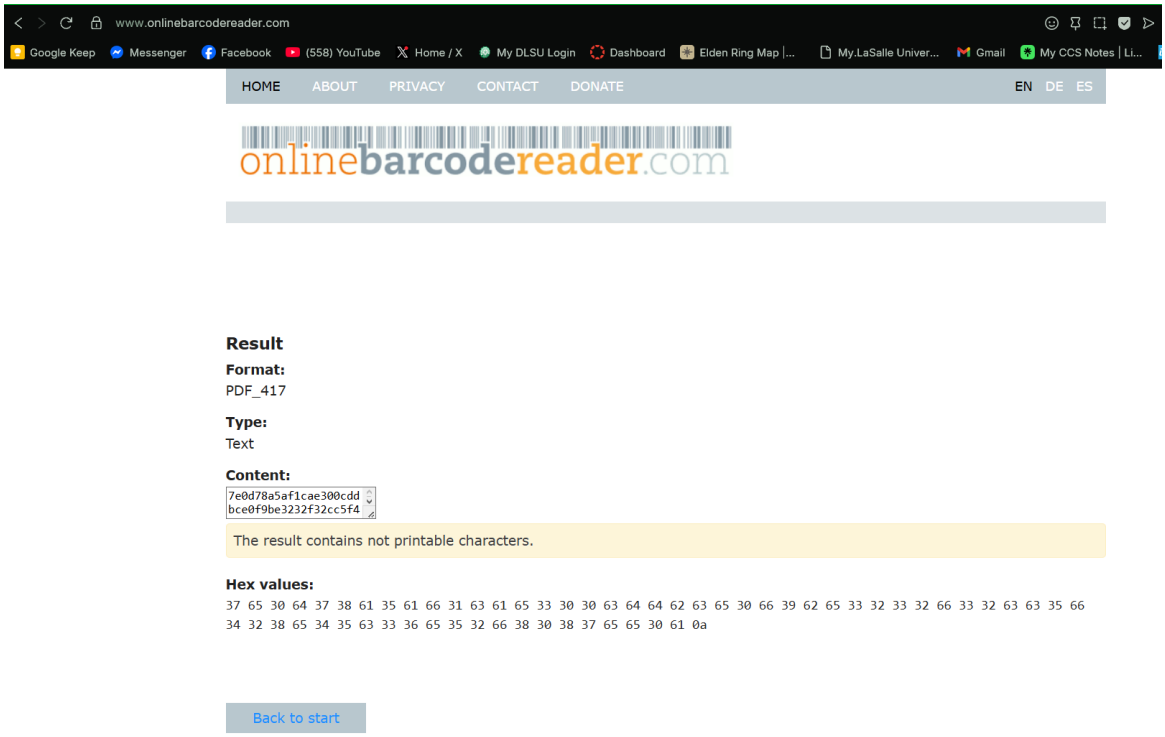


Figure x: Barcode scanned via onlinebarcodereader.com

The barcode image was then extracted and scanned using an online barcode decoder, which revealed a long hexadecimal string.

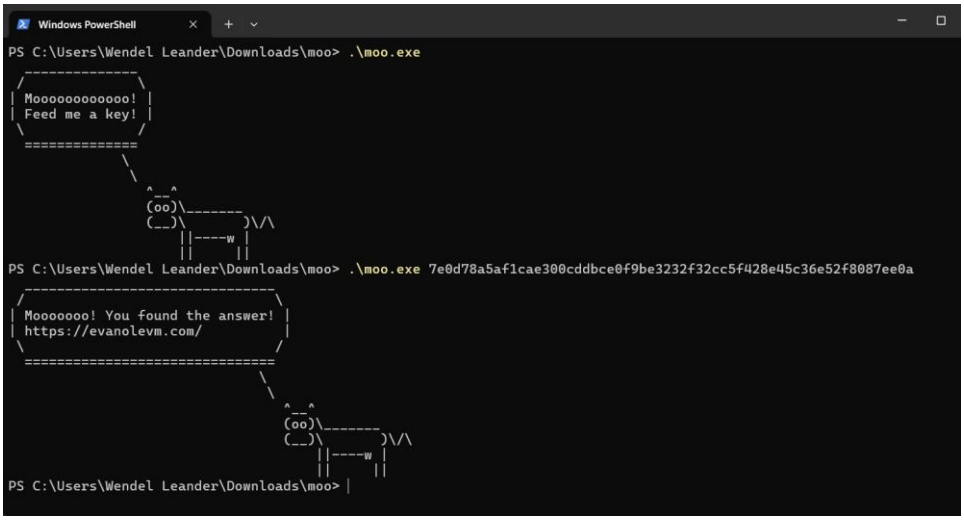


Figure x: moo.exe program revealing the flag

This hexadecimal string was then entered as the input key for the moo.exe program. Once submitted, the program returned the URL https://evanolevm.com/, which served as the flag for the challenge.

Flag: https://evanolevm.com/

Appendices

Appendix A: Collection of CTF Images and files

CTF images were downloaded from the official MVS 2026 images page: https://mvs2026.ctfd.io/images. CTF cipher files were downloaded from the official MVS 2026 challenges page: https://mvs2026.ctfd.io/challenges.

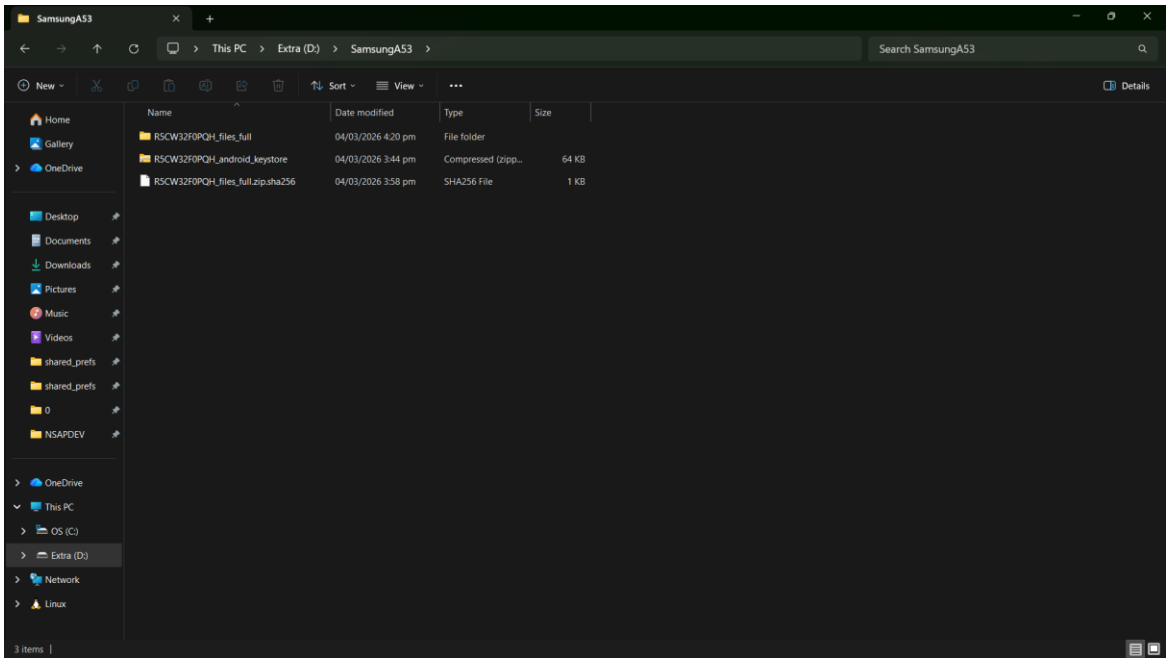


Figure x: Samsung A53 image downloaded and extracted

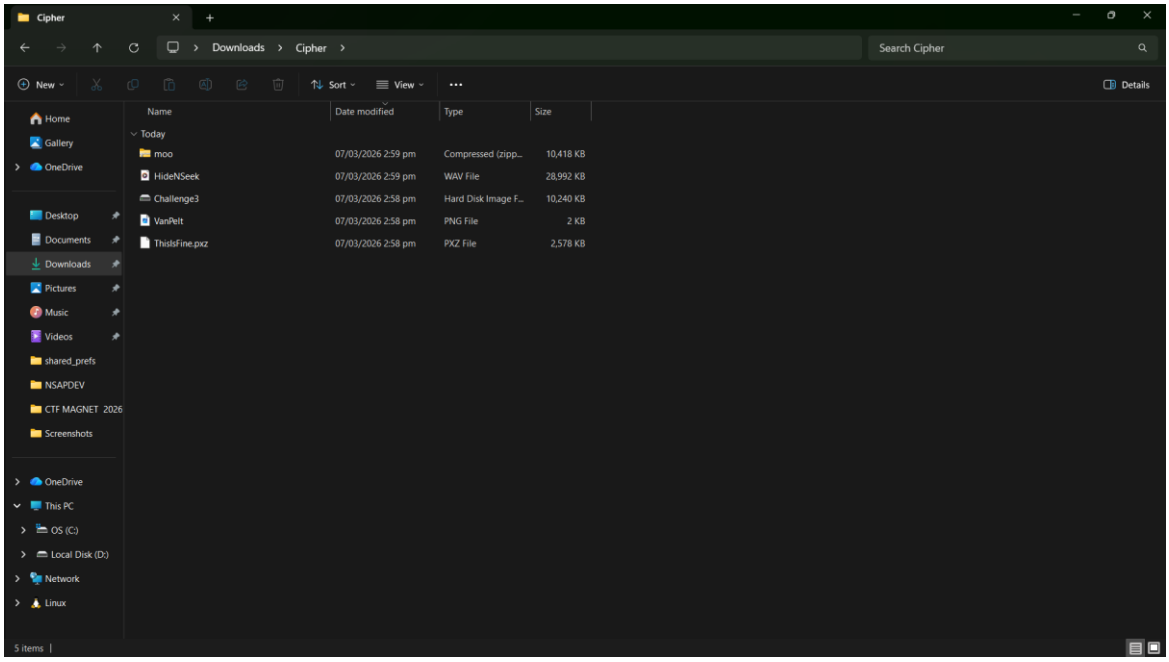


Figure x: Cipher files downloaded

Appendix B: Tools Used

This appendix lists the digital forensic and analysis tools used to identify, extract, and validate flags throughout the challenges. These tools supported artifact examination, database analysis, timestamp conversion, and file system inspection across Android, iOS, and cipher-based tasks.

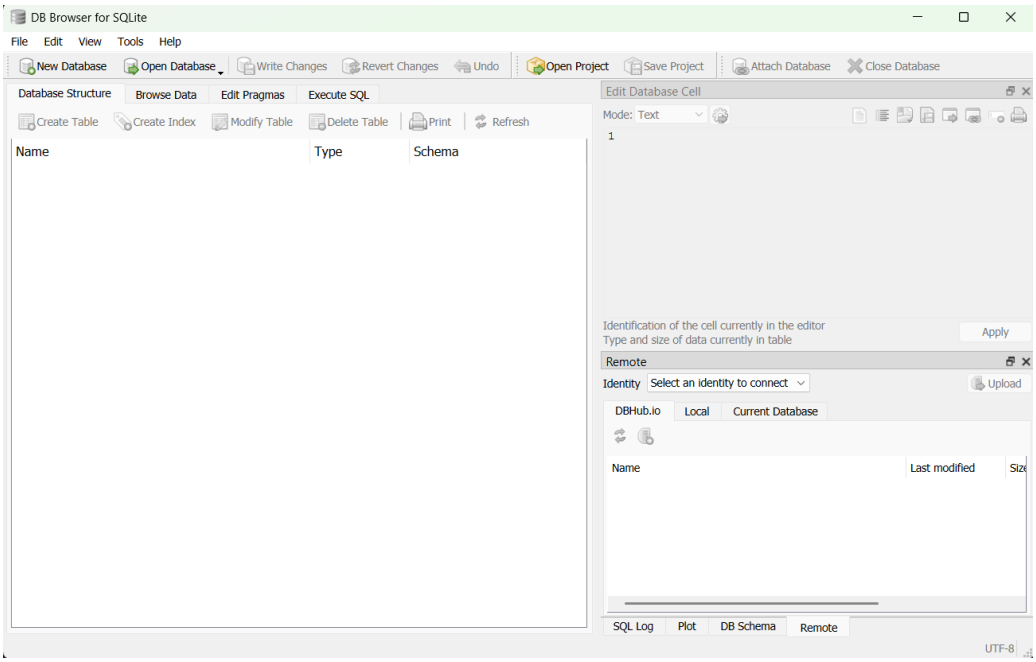


Figure x: DB Browser for SQLite used for viewing Databases

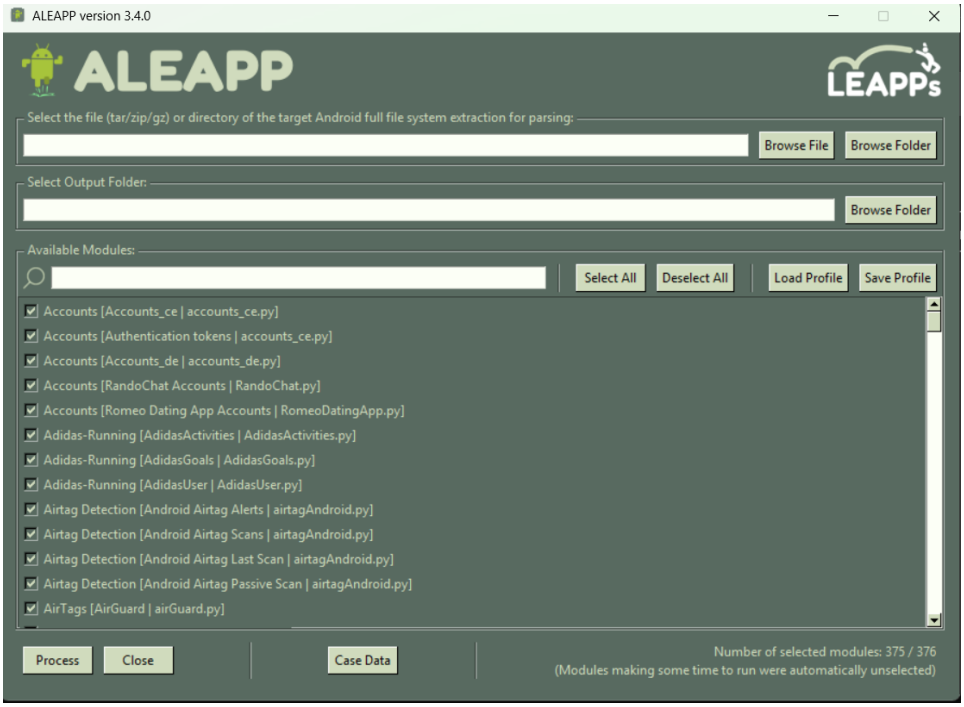


Figure x: ALEAPP used for Data Parsing

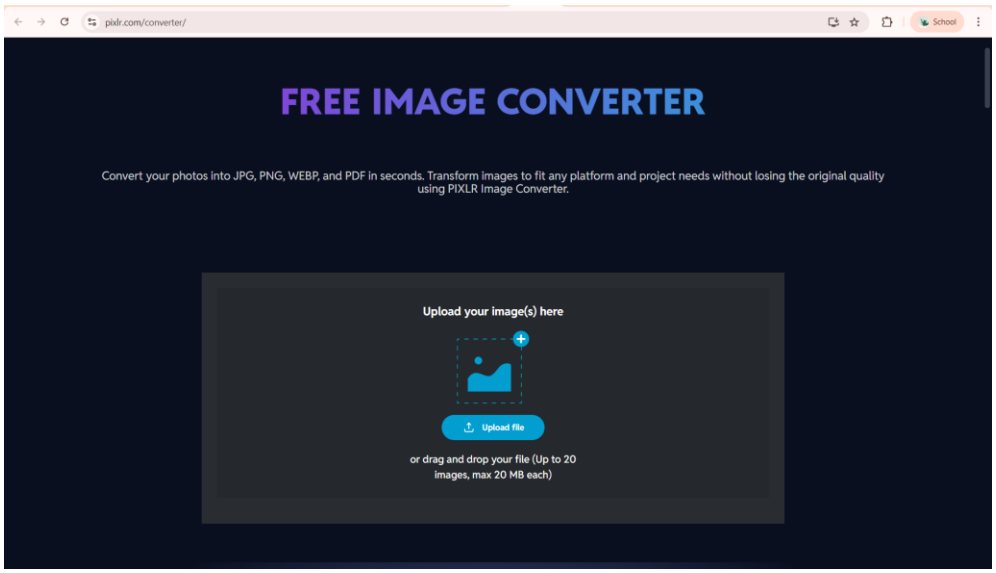


Figure x: Pixlr used for image conversion

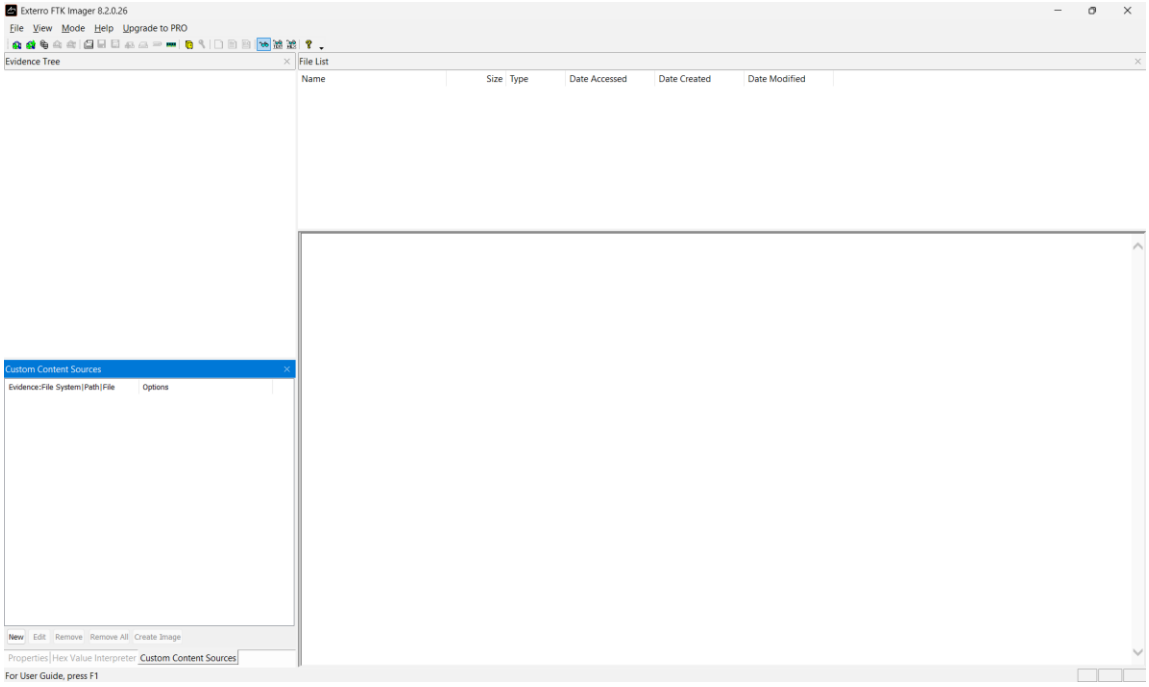


Figure x: FTK Imager used for checking hidden information in drives

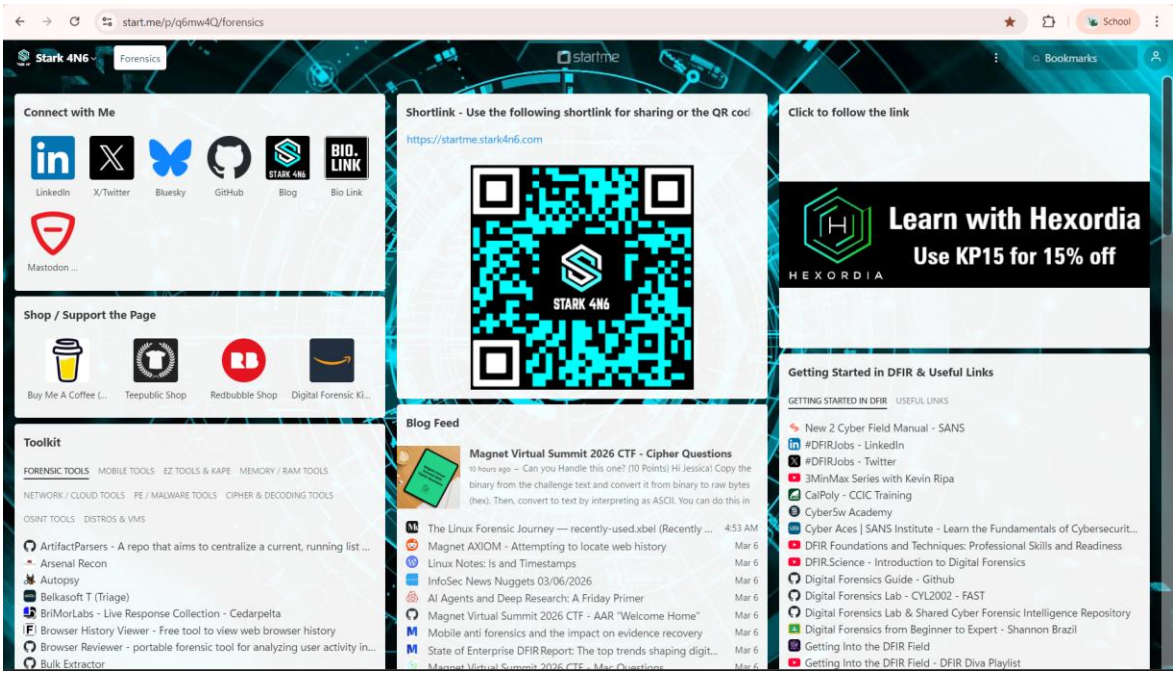


Figure x: Online Forensics Toolkit by stark4n6 used for finding usable tools

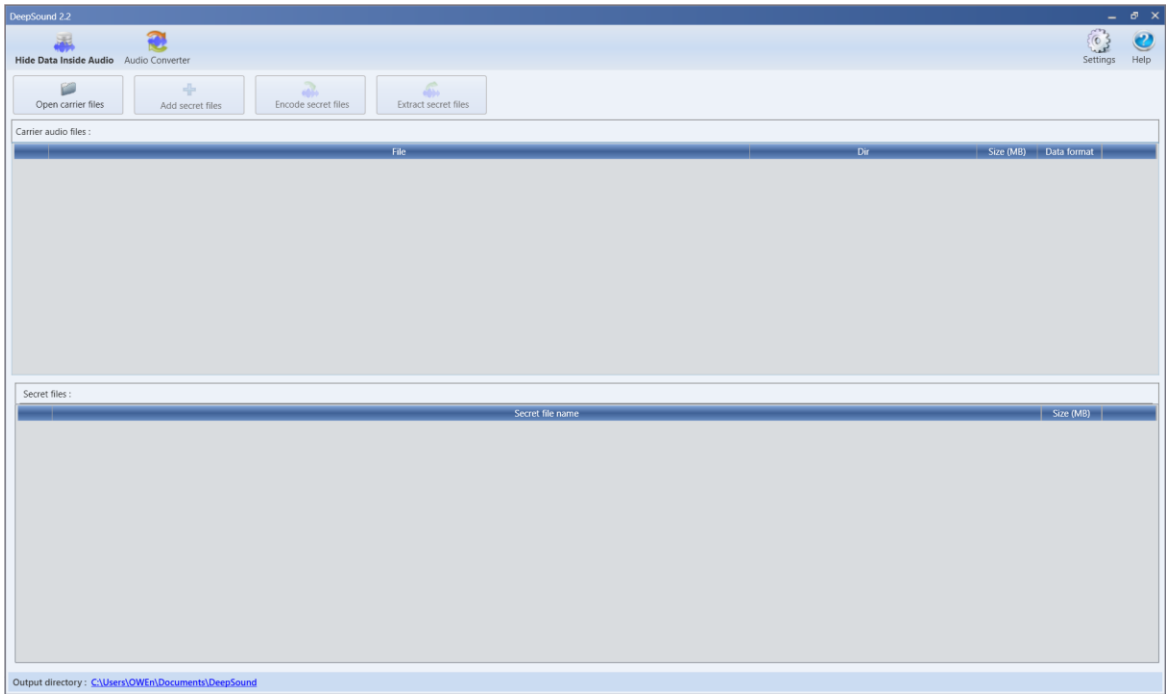


Figure x: DeepSound used for checking embedded data inside audio files

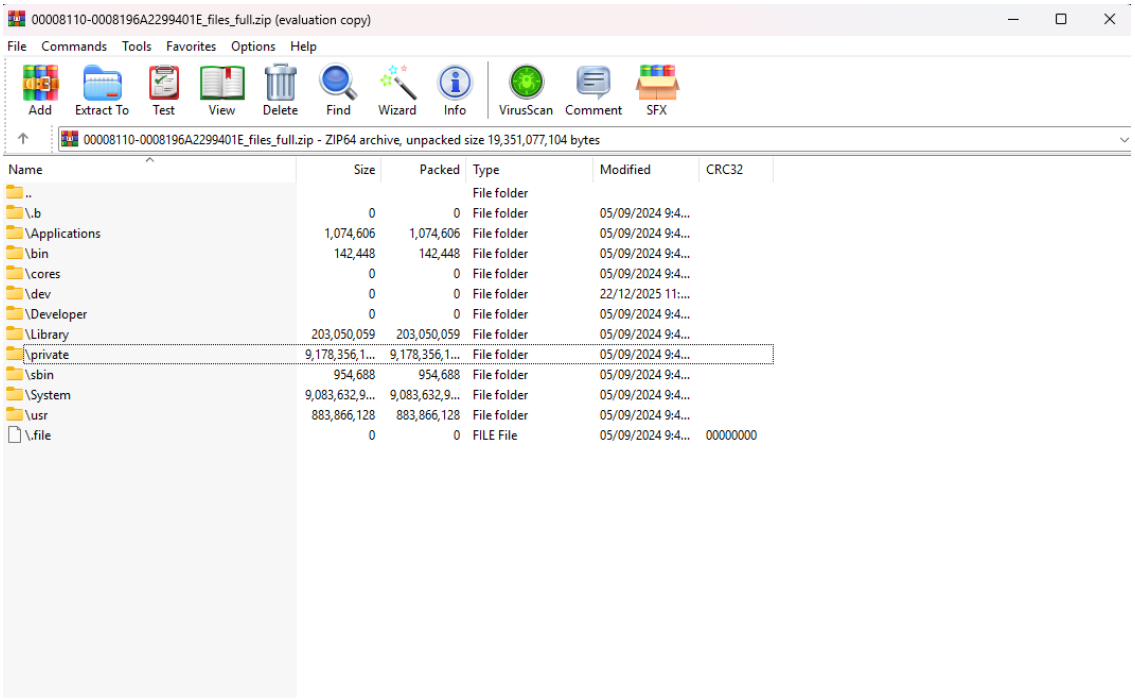


Figure x: WinRar used for locating specific files while compressed

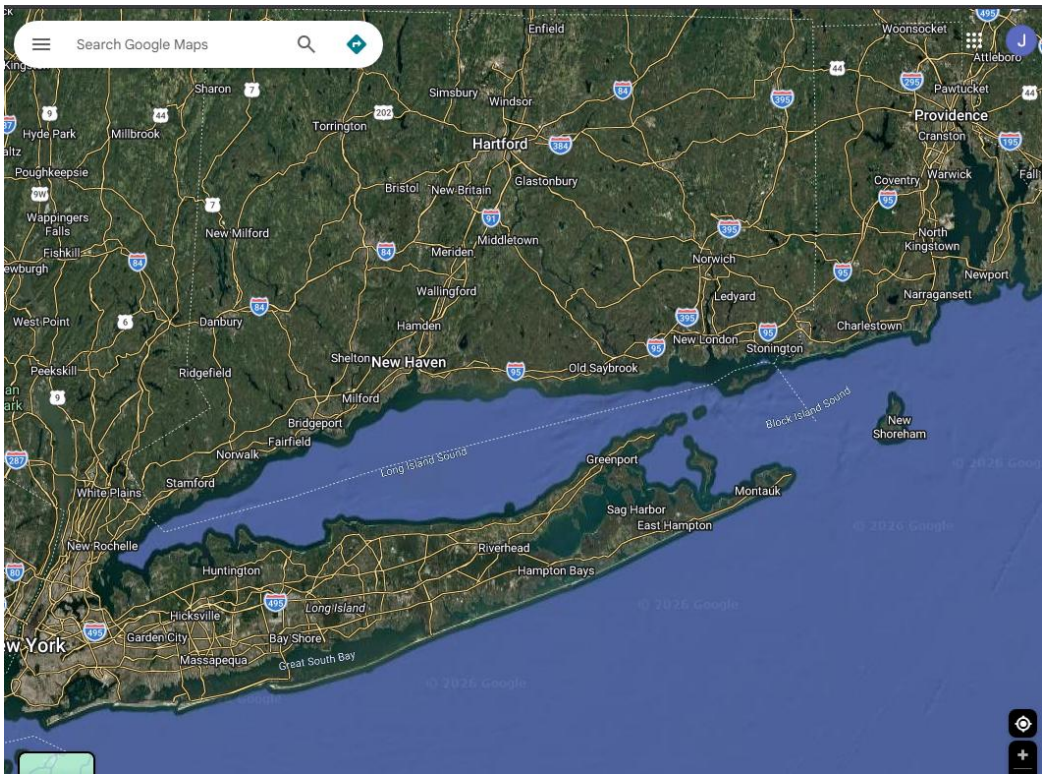


Figure x: Google Maps used for locating coordinates

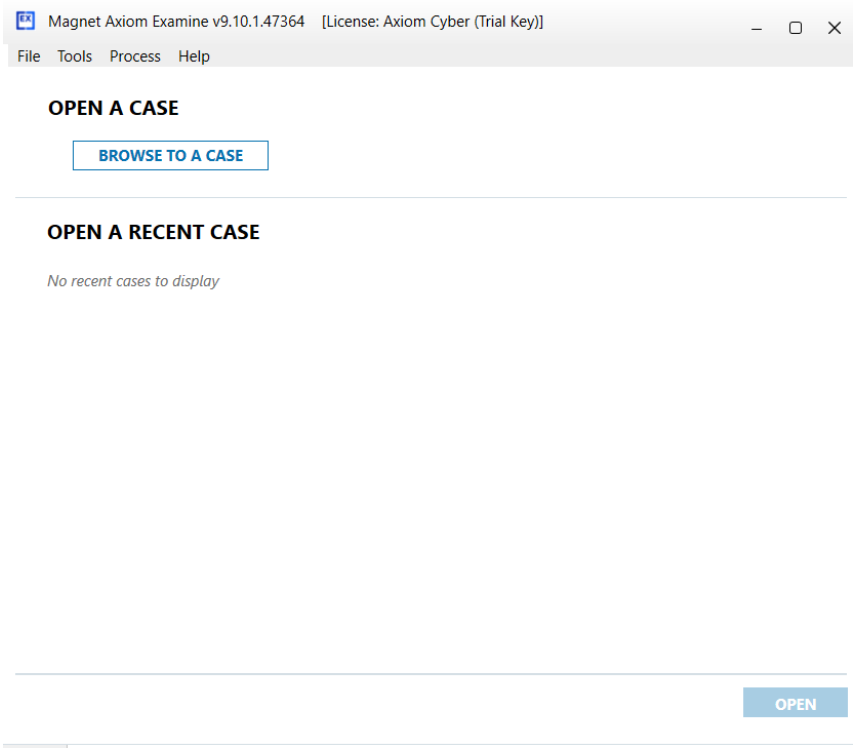
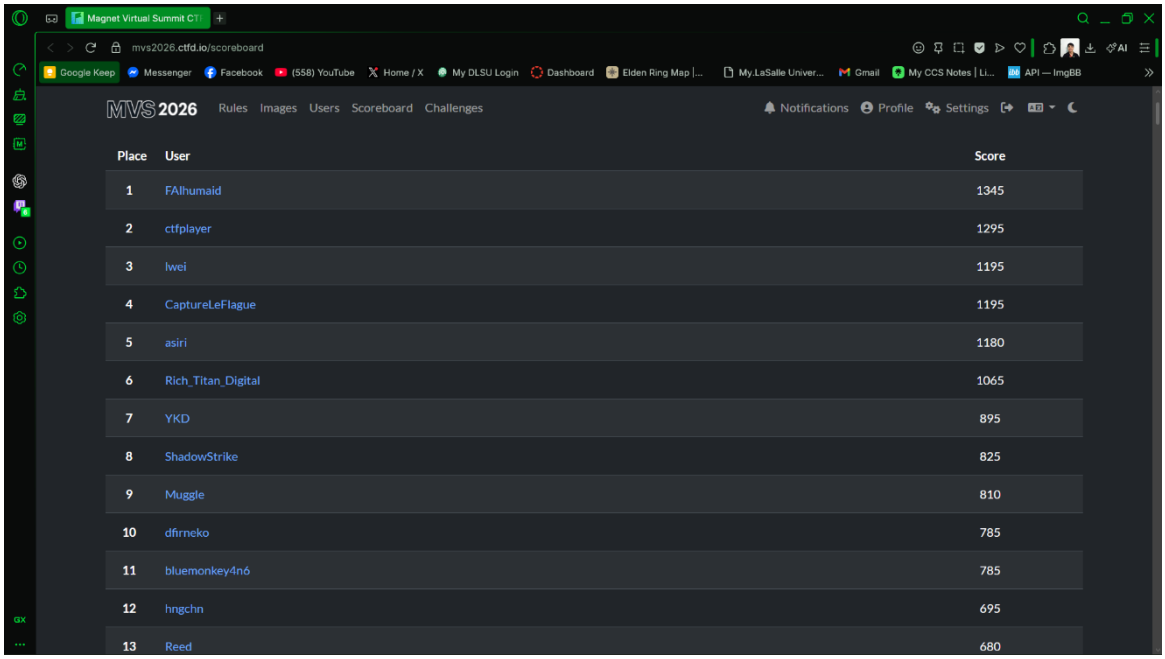


Figure x: Magnet Axiom

Appendix C: MVS 2026 CTF Group Rankings and Profiles

Ranks of accounts used in this CTF and the top leader board of MVS 2026 CTF.



Place	User	Score
1	FAlhumaid	1345
2	ctfplayer	1295
3	lwei	1195
4	CaptureLeFlague	1195
5	asiri	1180
6	Rich_Titan_Digital	1065
7	YKD	895
8	ShadowStrike	825
9	Muggle	810
10	dfirneko	785
11	bluemonkey4n6	785
12	hngchn	695
13	Reed	680

Figure x: Top leader board of MVS 2026 CTF

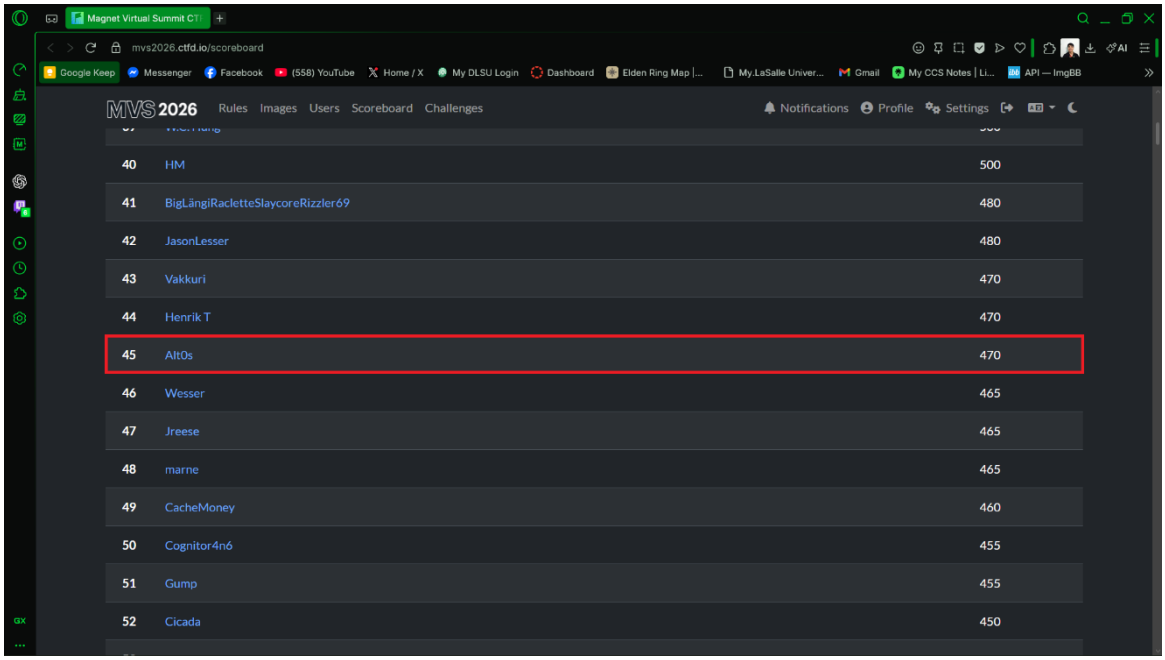


Figure x: NIS-S13-06 MVS 2026 CTF Group ranking

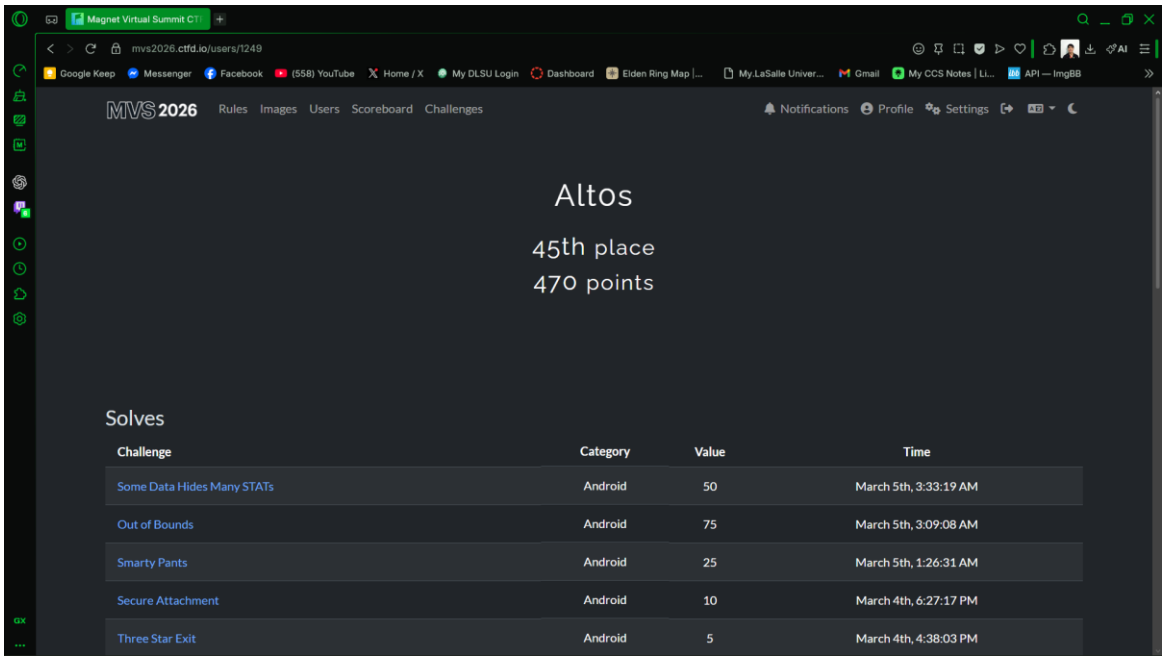


Figure x: MVS Group Account User Profile

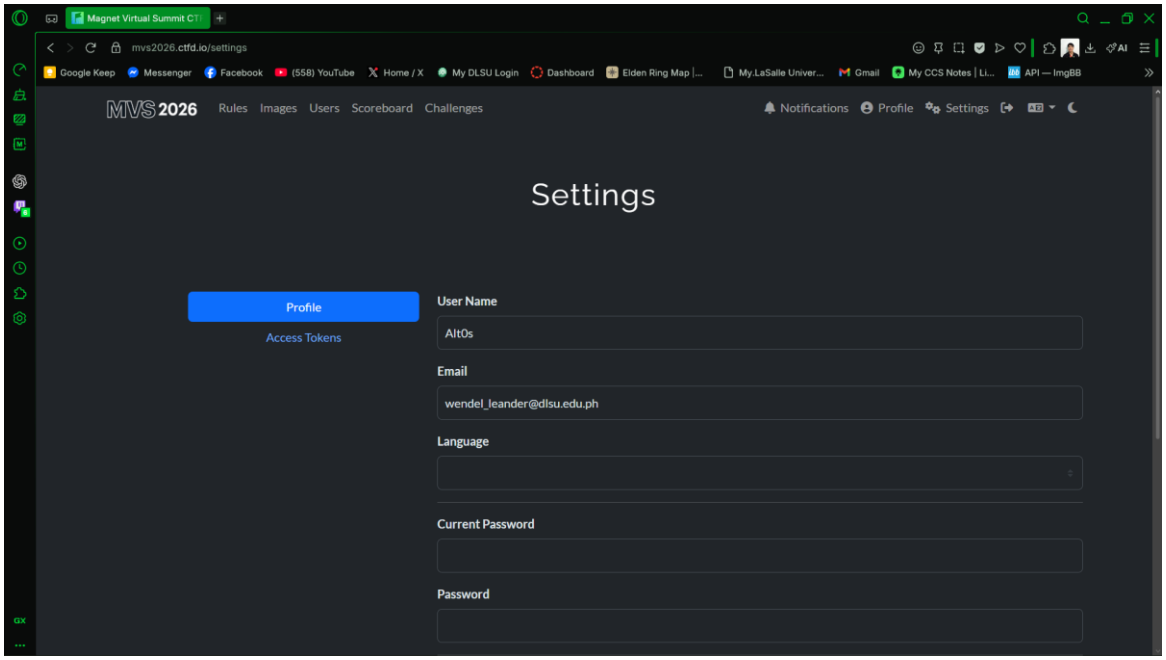


Figure x: MVS Group Account User Profile Settings

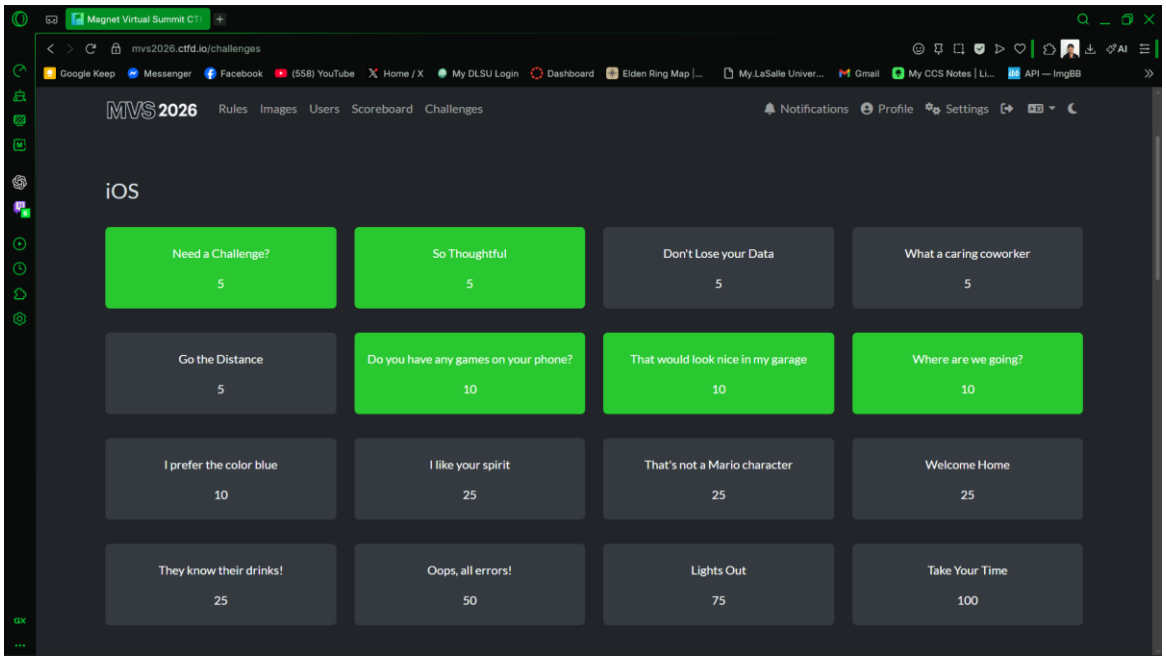


Figure x: MVS 2026 CTF Solved IOS Challenges

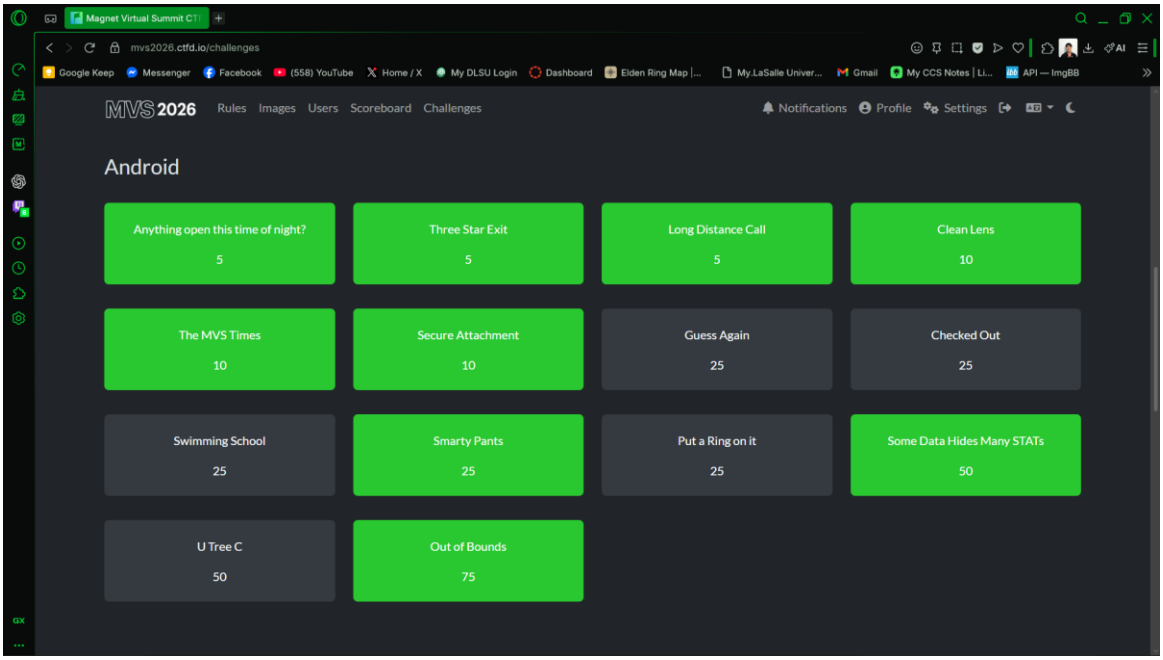


Figure x: MVS 2026 CTF Solved AdnroidChallenges

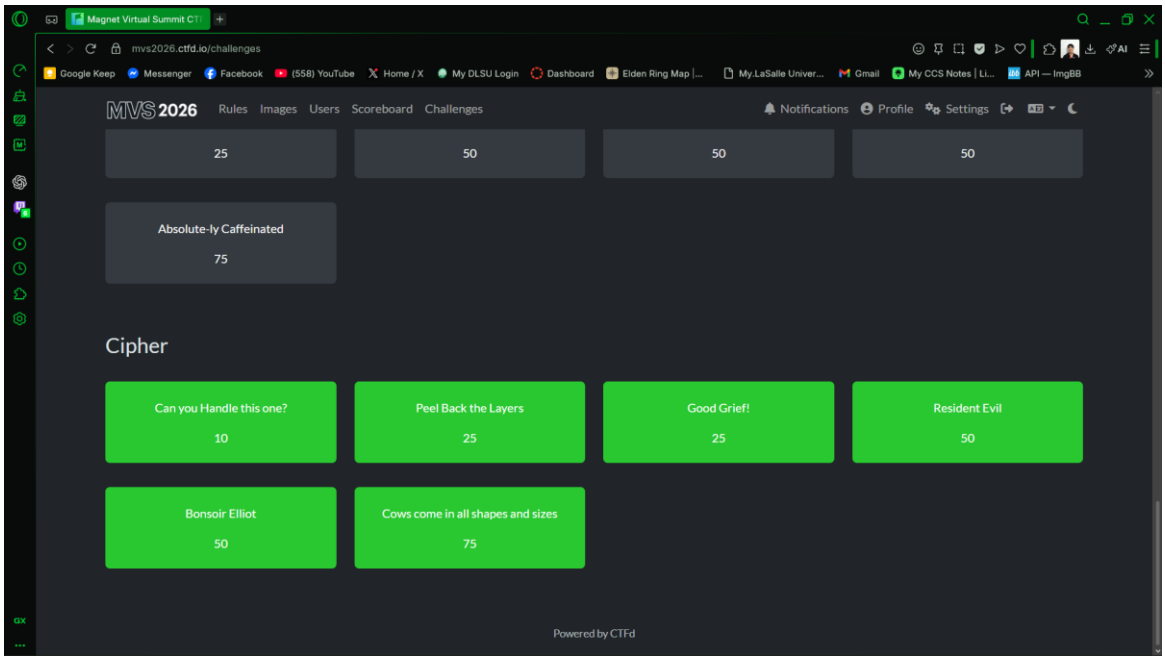


Figure x: MVS 2026 CTF Solved Cipher Challenges